

# Reducing burnout risk through sustainable work design in a hybrid, AI-enabled technology company

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## Group 11

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# Summary

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This project investigates how burnout risk can be reduced through sustainable work design in a hybrid, AI-enabled technology company. The case context shows that digital collaboration, deadline pressure, long working hours, and AI tool adoption may improve efficiency, but may also increase stress and burnout if job demands exceed job resources such as manager support and autonomy.

**Methodologically**, the project applies a mixed-method research design.

**The quantitative analysis** uses HRM survey data to validate key constructs through CFA and reliability checks, then applies regression, mediation, and moderation models to examine the drivers of burnout. The results show that deadline pressure affects burnout mainly through stress, while work hours remain a direct burnout driver. Manager support and autonomy reduce burnout directly, but they do not significantly buffer the deadline pressure–burnout relationship. Daily AI tool use is also associated with higher burnout, but it does not significantly amplify or buffer the effect of work demands.

**The qualitative phase** is designed as a follow-up study using semi-structured employee interviews, manager focus groups, and diary data to explain why these statistical patterns occur in daily work. Finally, the project evaluates whether Microsoft Viva Insights should replace annual surveys. The recommendation is not to replace surveys, but to combine surveys, Viva behavioral trace data, and targeted qualitative follow-ups to support evidence-based burnout prevention and sustainable performance.

# Strategic focus

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- **Strategic focus:**  
Reducing burnout risk through sustainable work design in a hybrid, AI-enabled technology company
- **Challenges:** The company needs to maintain high performance while preventing the work system from becoming unsustainable.

| Challenge               | Meaning                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------|
| High work demands       | Employees face deadline pressure, work hours, unexpected tasks, and changing priorities.                                  |
| Stress and burnout risk | If employees cannot recover from work pressure, stress may turn into burnout.                                             |
| Hybrid work complexity  | Remote/hybrid work gives flexibility, but may also blur work-life boundaries and increase digital collaboration overload. |
| AI-enabled work change  | AI tools may improve efficiency, but they may also increase expectations, speed, and pressure.                            |
| Managerial support gap  | Employees may need stronger support, prioritization, feedback, and barrier removal from managers.                         |
| Autonomy gap            | Employees may need more control over how they plan, prioritize, and perform their work.                                   |

**Why it matters:** Reducing burnout is valuable because it protects both **employee well-being** and **organizational performance**.

| Business value                  | Why it matters                                                                                  |
|---------------------------------|-------------------------------------------------------------------------------------------------|
| Sustainable performance         | Employees can maintain productivity without long-term exhaustion.                               |
| Better retention                | Lower burnout may reduce turnover intention and loss of knowledge.                              |
| Higher engagement               | Employees with better work-life balance are more likely to stay motivated.                      |
| Better collaboration            | Less stress can improve communication, teamwork, and coordination.                              |
| More innovation capacity        | Employees need focus time and mental energy to solve complex problems.                          |
| Better management decisions     | Analysis helps leaders identify whether pressure, support, autonomy, or work mode matters most. |
| Evidence-based HR interventions | The company can avoid generic well-being initiatives and target the real causes.                |

# Research questions

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## Main Research Question (business value)

What are the key drivers of employee burnout in a hybrid technology company, and to what extent do job resources — manager support and autonomy — buffer the relationship between job demands and burnout?

## Sub-Research Questions:

RQ1 Direct effects (regression): To what extent do deadline pressure and work hours predict employee burnout, controlling for individual and contextual characteristics?

RQ2 Mediation: Does occupational stress mediate the relationship between deadline pressure and burnout?

RQ3 Moderation : Does manager support moderate the relationship between deadline pressure and burnout, such that the effect is weaker for employees with higher managerial support?

RQ4 Secondary moderation: Does daily AI tool use moderate the relationship between work demands and burnout — either amplifying it (more cognitive load) or buffering it (efficiency gains)?

RQ5 Qualitative research: How do employees and managers explain the mechanisms through which deadline pressure and work hours create stress and burnout, and how do managerial support, autonomy, and AI tool use shape these experiences in daily work?

### Research Approach:

#### DEDUCTIVE →

Survey data, CFA, regression,  
mediation, moderation

#### INDUCTIVE →

Semi-structured interviews,  
qualitative coding

#### INTEGRATION

Quant findings drive  
interview focus + interpretation

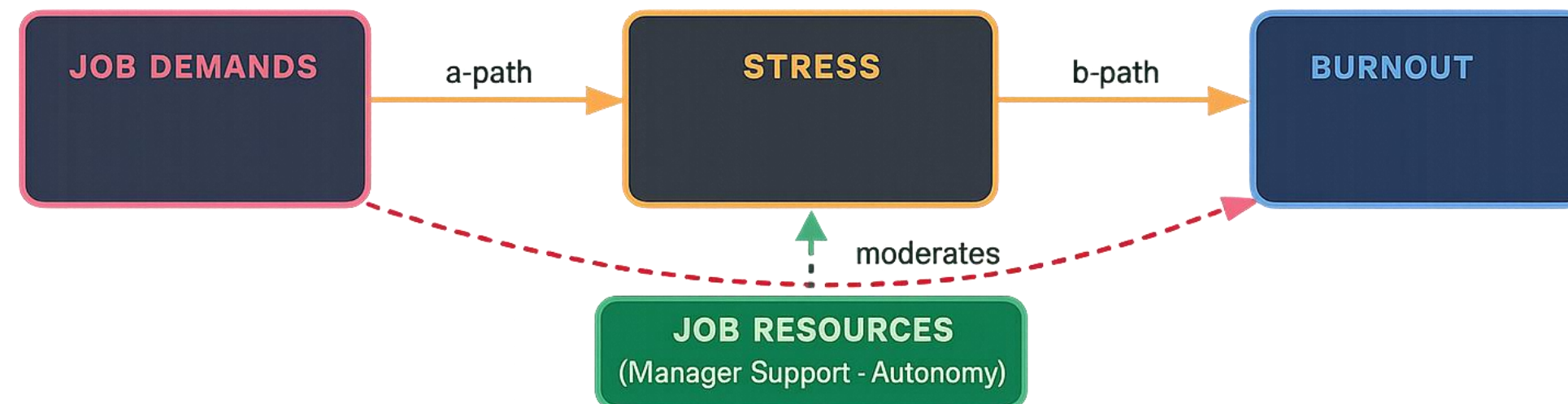
# Quantitative research

| Step | Quantitative analysis step             | Purpose                                                                                                                                                   |
|------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Data quality check                     | Check missing values, variable types, coding errors, impossible values, and whether categorical variables are correctly defined.                          |
| 3    | Descriptive statistics                 | Understand the basic pattern of the data, such as average work hours, burnout level, stress level, work mode distribution, and AI tool usage.             |
| 4    | Cronbach’s alpha reliability screening | Conduct an initial reliability check for each multi-item construct before CFA, such as burnout, stress, deadline pressure, manager support, and autonomy. |
| 5    | CFA measurement assessment             | Validate whether the survey items correctly measure the intended constructs and identify weak or conceptually misaligned items.                           |
| 6    | Omega / Composite Reliability check    | Assess final post-CFA reliability based on the confirmed measurement model. This provides stronger reliability evidence than Cronbach’s alpha alone.      |
| 7    | Create final construct scores          | Create composite scores or factor scores for burnout, stress, deadline pressure, manager support, and autonomy based on retained items.                   |
| 8    | Correlation matrix                     | Examine initial relationships between key constructs and identify possible multicollinearity before regression analysis.                                  |

# Quantitative research

| Step | Quantitative analysis step         | Purpose                                                                                                                                   |
|------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 9    | Baseline regression                | Test the basic relationship between deadline pressure and burnout while including essential control variables.                            |
| 10   | Extended regression                | Add stress, manager support, and autonomy to examine whether job demands and job resources explain burnout more fully.                    |
| 11   | Mediation analysis                 | Test whether stress explains the relationship between deadline pressure and burnout.                                                      |
| 12   | Moderation analysis 1              | Test whether manager support weakens the positive relationship between deadline pressure and burnout.                                     |
| 13   | Moderation analysis 2              | Test whether autonomy weakens the positive relationship between deadline pressure and burnout.                                            |
| 14   | Diagnostics and assumptions        | Check model quality through diagnostics such as multicollinearity, residual patterns, and influential observations.                       |
| 15   | Interpretation and recommendations | Translate statistical findings into practical management recommendations for reducing burnout risk and improving sustainable work design. |

**Main focus:  
(Step 11-13)**



# Quantitative Data - Review and quality check

## Data Descriptive

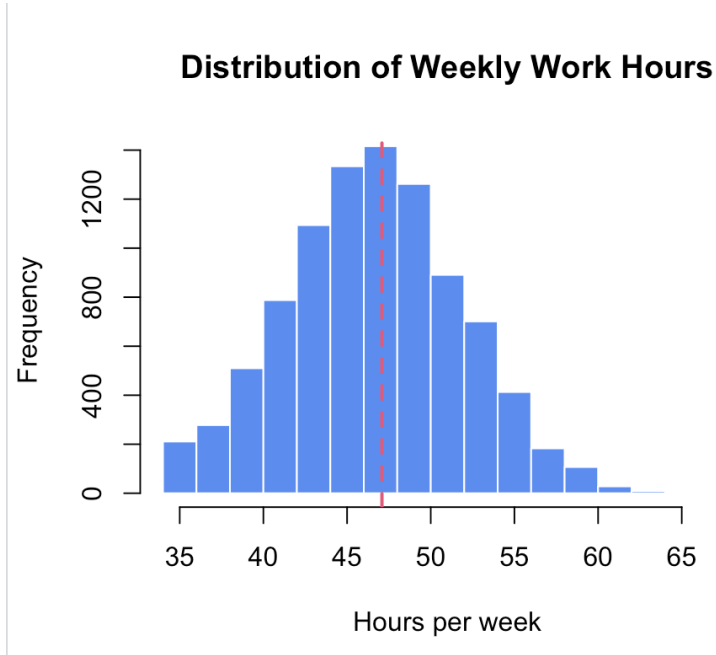
|                     |      |
|---------------------|------|
| Columns (variables) | 33   |
| Rows(size)          | 9230 |
| Scale type          | 1–7  |

## Data quality and summary

| Indicator      | Result              | Interpretation                                                                                              |
|----------------|---------------------|-------------------------------------------------------------------------------------------------------------|
| Missing values | 0                   | excellent data completeness                                                                                 |
| Skewness       | Close to 0          | The construct distributions are approximately symmetric, so there is no serious skewness issue.             |
| Kurtosis       | Around -0.8 to -1.0 | The distributions are slightly flatter than normal, but not problematic for regression with a large sample. |

## Data Distribution

| Key finding                                                                 | Interpretation                                                                             |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Average work hours = <b>47.1 hours/week</b>                                 | Indicates a potential overwork pattern and supports the focus on sustainable work design.  |
| Average burnout $\approx$ <b>4.0 / 7</b>                                    | Burnout is not extreme, but it is clearly present around the scale midpoint.               |
| Remote burnout = <b>4.05</b> , Hybrid = <b>4.00</b> , On-site = <b>3.91</b> | Remote and hybrid employees show slightly higher burnout, suggesting work mode may matter. |
| AI users burnout = <b>4.04</b> , non-users = <b>3.88</b>                    | AI users show slightly higher burnout, but this may reflect workload or role differences.  |
| Mean tenure = <b>2.5 years</b>                                              | The workforce is relatively new, so support and onboarding may be important.               |



**Work hours range from 35 to 68 hours per week, with an average of 47.1 hours, suggesting a potential overwork pattern.**

# Quantitative Data - CFA

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**We choose CFA instead EFA.**

Reason:

Our survey constructs are **not exploratory**. They are already:

- theoretically defined in the HRM/work design literature
- measured through pre-designed multi-item survey scales
- grouped in the questionnaire by construct: **Burnout, Stress, Deadline Pressure, Manager Support, and Autonomy**

**Purpose of CFA:**

Confirmatory Factor Analysis was conducted to validate whether the survey items correctly measure the intended latent constructs:

- **Burnout**
- **Stress**
- **Deadline Pressure**
- **Manager Support**
- **Autonomy**

CFA helps us check whether each item loads strongly on its intended construct and whether any items should be excluded due to weak loading or conceptual mismatch.

# Quantitative Data - Reliability-Pre CFA

**Data reliability– Alpha** Higher Cronbach’s Alpha values indicate greater internal consistency among indicators.

| KPI                                                        | Result                                                                                          | Threshold                         | Interpretation                                                                              |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------|
| <b>Cronbach’s alpha</b>                                    | Burnout = 0.804; Stress = 0.776; Deadline = 0.732;<br>Manager support = 0.750; Autonomy = 0.758 | > 0.70                            | All constructs show acceptable pre-CFA internal consistency.                                |
| <b>Removed items<br/>(alpha if delete in<br/>appendix)</b> | BURN4, STR5, DEAD2, MSUP3, AUTO1                                                                | Based on theory + alpha/CFA       | Items were removed because they were conceptually less aligned with the intended construct. |
| <b>CFA model fit</b>                                       | CFI = 1.000; TLI = 1.000; RMSEA = 0.000; SRMR = 0.007                                           | CFI/TLI > 0.90; RMSEA/SRMR < 0.08 | Excellent measurement model fit.                                                            |
| <b>Standardized loadings</b>                               | Approx. 0.661–0.723                                                                             | > 0.50 acceptable; > 0.70 ideal   | All retained items load acceptably on their intended construct.                             |
| <b>Omega reliability</b>                                   | Burnout = 0.793; Stress = 0.798; Deadline = 0.770;<br>Manager support = 0.776; Autonomy = 0.791 | > 0.70                            | Post-CFA reliability is acceptable for all constructs.                                      |

All constructs exceed the commonly used threshold of 0.70, indicating acceptable internal consistency and supporting further measurement validation through CFA.

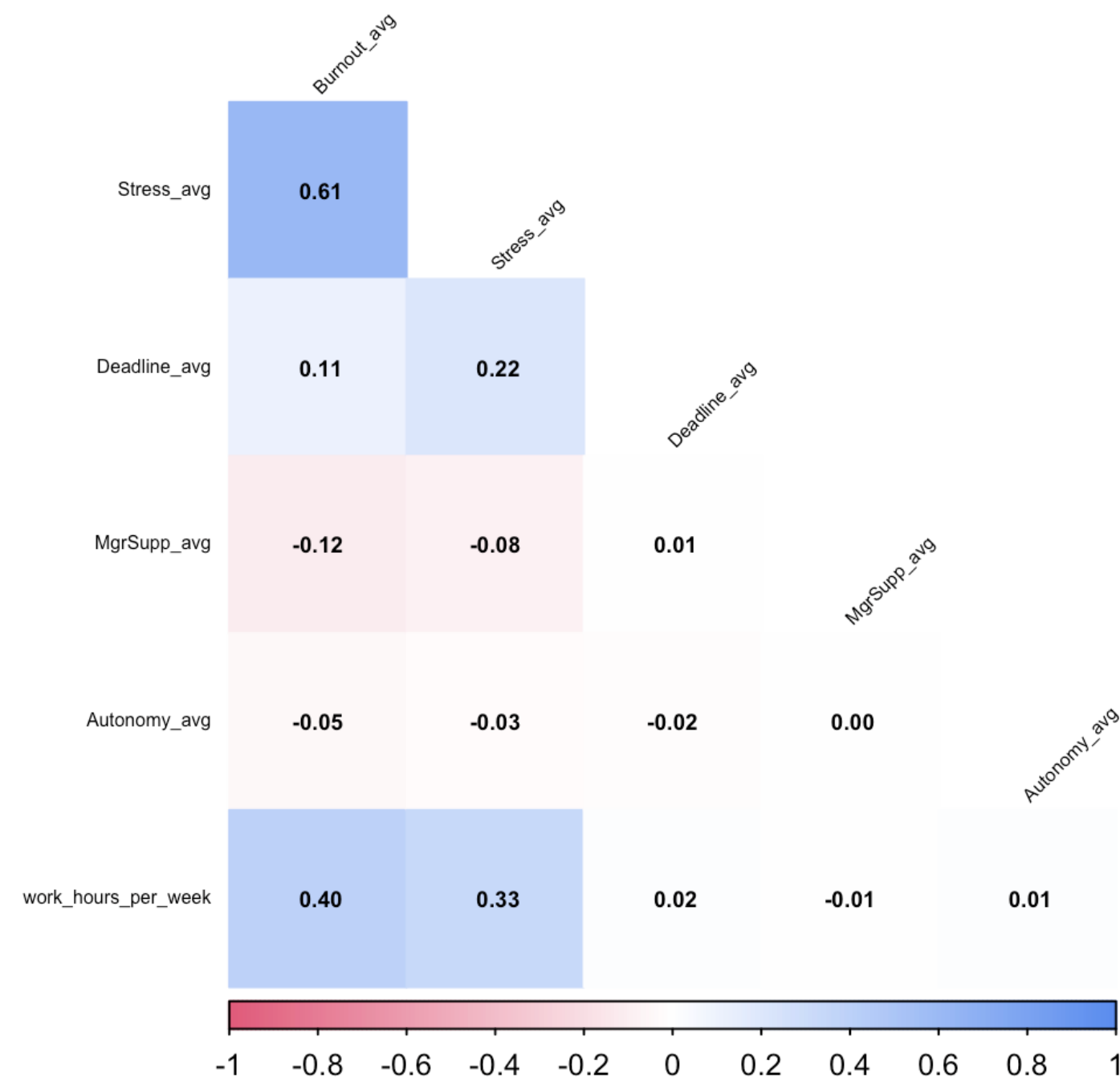
Based on both statistical and conceptual reasoning, We removed five weaker items: BURN4, STR5, DEAD2, MSUP3, and AUTO1.

The final CFA model shows excellent fit, with CFI and TLI equal to 1.00, RMSEA equal to .000, and SRMR equal to .007.

All retained factor loadings are above **0.50**, and omega reliability is above **0.70** for all constructs.

Therefore, the measurement model is suitable for further analysis.

# Quantitative Data - Correlation matrix: initial relationship pattern



**The correlation matrix shows that burnout is most strongly related to stress and weekly work hours.**

Deadline pressure is positively related to stress and burnout, while manager support and autonomy are negatively related to burnout.

These patterns provide preliminary support for the proposed burnout pathway, but regression and mediation/moderation models are needed for hypothesis testing.

## Quantitative Data - Direct effects (regression)

### Model 1: Baseline regression

Burnout = deadline pressure + work hours + controls  
Purpose: test whether work demands predict burnout.

| KPI                     | Result       | Interpretation                                      |
|-------------------------|--------------|-----------------------------------------------------|
| Adjusted R <sup>2</sup> | <b>0.188</b> | Baseline model explains 18.8% of burnout variation. |
| F-test                  | p < .001     | The model is statistically significant.             |
| VIF                     | All ≈ 1      | No multicollinearity issue.                         |

### Model 2: Extended regression

Burnout = deadline pressure + stress + manager support + autonomy + work hours + controls  
Purpose: see whether stress explains part of the relationship and whether job resources reduce burnout.

| KPI                     | Result       | Interpretation                                      |
|-------------------------|--------------|-----------------------------------------------------|
| Adjusted R <sup>2</sup> | <b>0.424</b> | Extended model explains 42.4% of burnout variation. |
| F-test                  | p < .001     | The model is statistically significant.             |
| VIF                     | 1.00–1.25    | No serious multicollinearity concern.               |

## Model 1: Baseline regression

| Predictor          | Estimate      | Interpretation                                                          |
|--------------------|---------------|-------------------------------------------------------------------------|
| Deadline pressure  | <b>+0.108</b> | Higher deadline pressure is associated with higher burnout.             |
| Work hours/week    | <b>+0.108</b> | Longer working hours are associated with higher burnout.                |
| Hierarchical level | <b>+0.140</b> | Higher-level employees report higher burnout.                           |
| Hybrid work mode   | <b>+0.106</b> | Hybrid employees report slightly higher burnout than on-site employees. |
| Remote work mode   | <b>+0.161</b> | Remote employees report higher burnout than on-site employees.          |
| Daily AI tool use  | <b>+0.147</b> | Daily AI users report higher burnout than non-users.                    |
| Age                | -0.000        | No meaningful relationship with burnout.                                |
| Tenure             | -0.003        | No meaningful relationship with burnout.                                |

The baseline model shows that deadline pressure, work hours, remote/hybrid work, hierarchy level, and daily AI tool use are associated with higher burnout.

This supports the strategic focus on work demands and sustainable work design.

Model 2: Extended regression

| Predictor          | Estimate | Interpretation                                                                  |
|--------------------|----------|---------------------------------------------------------------------------------|
| Stress             | +0.519   | Stress is the strongest predictor of burnout.                                   |
| Work hours/week    | +0.065   | Long working hours still predict burnout after adding stress and resources.     |
| Manager support    | -0.075   | Higher manager support is associated with lower burnout.                        |
| Autonomy           | -0.043   | Higher autonomy is associated with lower burnout.                               |
| Deadline pressure  | -0.006   | Deadline pressure no longer directly predicts burnout after stress is included. |
| Hierarchical level | +0.033   | Seniority still has a small positive association with burnout.                  |
| Hybrid work mode   | +0.080   | Hybrid employees remain slightly higher than on-site employees.                 |
| Remote work mode   | +0.114   | Remote employees remain higher than on-site employees.                          |
| Daily AI tool use  | +0.098   | Daily AI users remain slightly higher in burnout.                               |

After adding stress, manager support, and autonomy, the model explains much more burnout variation. Stress becomes the central mechanism, while manager support and autonomy act as protective job resources. The disappearance of the deadline pressure effect suggests that deadline pressure may influence burnout mainly through stress.

# Quantitative Data - Mediation and Moderation

## Hypothesis 1

Deadline pressure is positively associated with burnout, and this relationship is mediated by employee stress.



## 01. Research design

| Role                            | Variable                                                                        |
|---------------------------------|---------------------------------------------------------------------------------|
| <b>X / Independent variable</b> | Deadline pressure                                                               |
| <b>M / Mediator</b>             | Stress                                                                          |
| <b>Y / Dependent variable</b>   | Burnout                                                                         |
| <b>Controls</b>                 | Work hours per week, work mode, AI tools daily, age, tenure, hierarchical level |

## Test

### Path a:

Deadline pressure → Stress

### Path b:

Stress → Burnout

### Direct effect c':

Deadline pressure → Burnout, controlling for stress

### Indirect effect a × b:

Deadline pressure → Stress → Burnout

## Hypothesis 1

Deadline pressure is positively associated with burnout, and this relationship is mediated by employee stress.

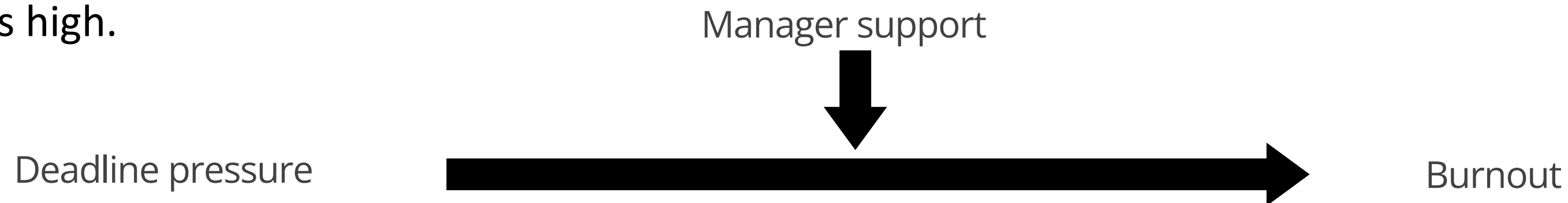


| Effect / path                                                       | $\beta$ | p-value | 95% CI          | Interpretation                                                                                                                     |
|---------------------------------------------------------------------|---------|---------|-----------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>a-path:</b> Deadline pressure → Stress                           | 0.219   | < 0.001 | [0.201, 0.238]  | Deadline pressure significantly predicts stress. For every one-unit increase in deadline pressure, stress increases by .219 units. |
| <b>b-path:</b> Stress → Burnout                                     | 0.530   | < 0.001 | [0.513, 0.547]  | Stress significantly predicts burnout and is the strongest path in the model.                                                      |
| <b>c'-path:</b> Deadline pressure → Burnout, controlling for stress | -0.007  | 0.372   | [-0.024, 0.009] | The direct effect is non-significant once stress is included. This is the key finding.                                             |
| <b>Indirect effect:</b> Deadline pressure → Stress → Burnout        | 0.116   | < 0.001 | [0.106, 0.127]  | The indirect effect is significant and the confidence interval excludes zero, confirming mediation.                                |
| <b>Direct effect</b>                                                | -0.007  | 0.372   | [-0.024, 0.009] | Same as the c'-path: deadline pressure no longer directly predicts burnout after stress is included.                               |
| <b>Total effect</b>                                                 | 0.109   | < 0.001 | [0.090, 0.127]  | Deadline pressure has a significant overall effect on burnout before separating the stress pathway.                                |

**Full mediation confirmed** Deadline pressure does not directly predict burnout ( $\beta = -.007$ ,  $p = .372$ ) once stress is included. Instead, the entire effect operates through stress accumulation (indirect effect  $\beta = .116$ , 95% CI [.106, .127]). This indicates full mediation support hypothesis 1, Deadline pressure cause Stress then cause Burnout

## Hypothesis 2

Manager support moderates the relationship between deadline pressure and burnout, such that the positive relationship between deadline pressure and burnout is weaker when manager support is high.



## 01. Research design

$$\text{Burnout} = \beta_0 + \beta_1 \text{DeadlinePressure} + \beta_2 \text{ManagerSupport} + \beta_3 \text{DeadlinePressure} \times \text{ManagerSupport} + \text{Controls} + \varepsilon$$

| Role                            | Variable                                                                        |
|---------------------------------|---------------------------------------------------------------------------------|
| <b>X / Independent variable</b> | Deadline pressure                                                               |
| <b>Y / Dependent variable</b>   | Burnout                                                                         |
| <b>Moderator / W</b>            | Manager support                                                                 |
| <b>Controls</b>                 | Work hours per week, work mode, AI tools daily, age, tenure, hierarchical level |

### - Interaction effect $\beta_3$ :

Deadline pressure  $\times$  Manager support  $\rightarrow$  Burnout

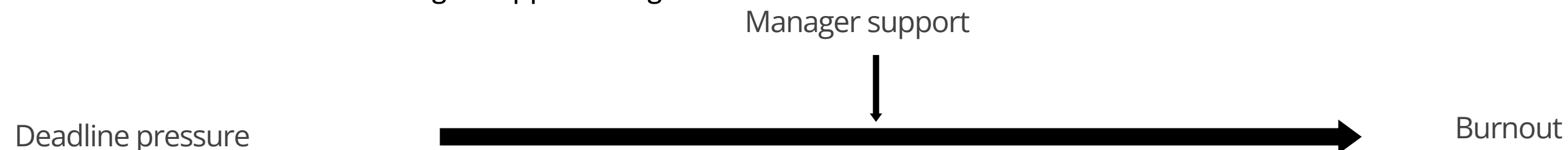
Checks whether manager support changes the strength of the deadline pressure–burnout relationship.

# Group 11 Quantitative Data - Mediation and Moderation

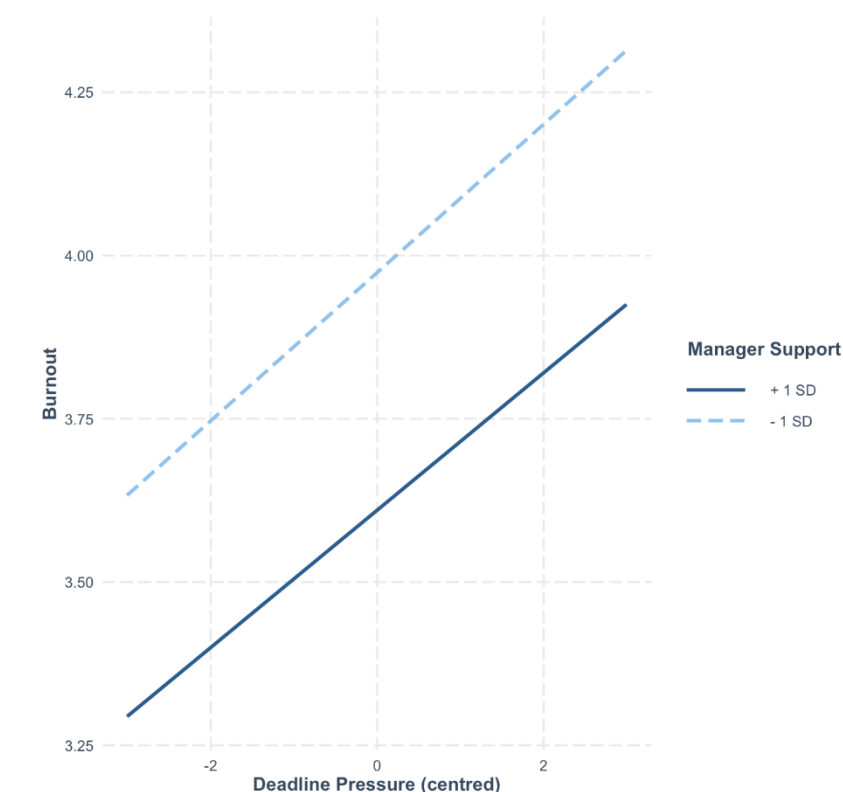
## Hypothesis 2

Research Proposal Presentation

Manager support moderates the relationship between deadline pressure and burnout, such that the positive relationship between deadline pressure and burnout is weaker when manager support is high.



| Test                                             | Key result                              | Interpretation                                            |
|--------------------------------------------------|-----------------------------------------|-----------------------------------------------------------|
| Interaction: Deadline pressure × Manager support | $\beta = -0.003, p = .658$              | Not significant. H2 is not supported.                     |
| Manager support main effect                      | $\beta = -0.118, p < .001$              | Manager support directly reduces burnout.                 |
| R <sup>2</sup> test                              | $F = 0.196, p = .658$                   | Adding the interaction does not improve the model.        |
| Simple slopes                                    | Low support = .114; High support = .105 | Slopes are almost identical, meaning no buffering effect. |

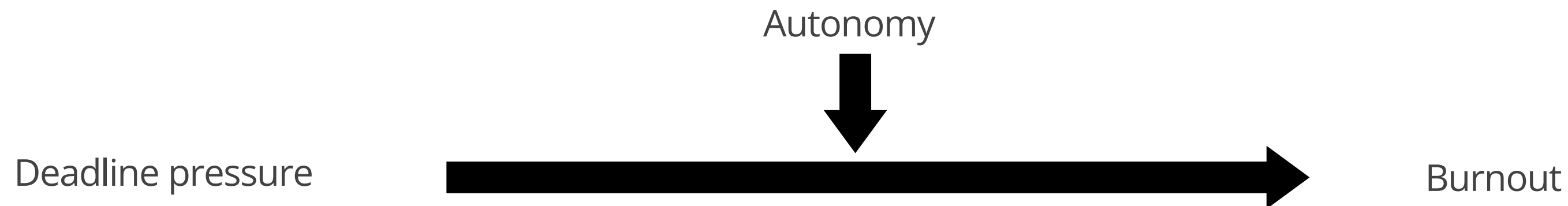


Manager support acts as a direct buffer, not a moderator The interaction term was non-significant ( $\beta = -.003, p = .658, R^2 \approx .000$ ). Manager support does not weaken the deadline–burnout relationship. However, manager support has a significant negative main effect on burnout ( $\beta = -.118, p < .001$ ) — meaning employees with higher manager support report lower burnout overall, regardless of deadline pressure level.

Supportive managers reduce overall burnout levels, but they do not significantly change how strongly deadline pressure translates into burnout.

### Hypothesis 3

Autonomy moderates the relationship between deadline pressure and burnout, such that the positive relationship between deadline pressure and burnout is weaker when employee autonomy is high.



## 01. Research design

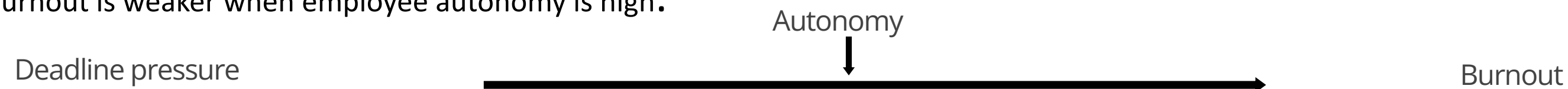
$$\text{Burnout} = \beta_0 + \beta_1 \text{DeadlinePressure} + \beta_2 \text{Autonomy} + \beta_3 \text{DeadlinePressure} \times \text{Autonomy} + \text{Controls} + \varepsilon$$

| Role                            | Variable                                                                        |
|---------------------------------|---------------------------------------------------------------------------------|
| <b>X / Independent variable</b> | Deadline pressure                                                               |
| <b>Y / Dependent variable</b>   | Burnout                                                                         |
| <b>Moderator / W</b>            | Autonomy                                                                        |
| <b>Controls</b>                 | Work hours per week, work mode, AI tools daily, age, tenure, hierarchical level |

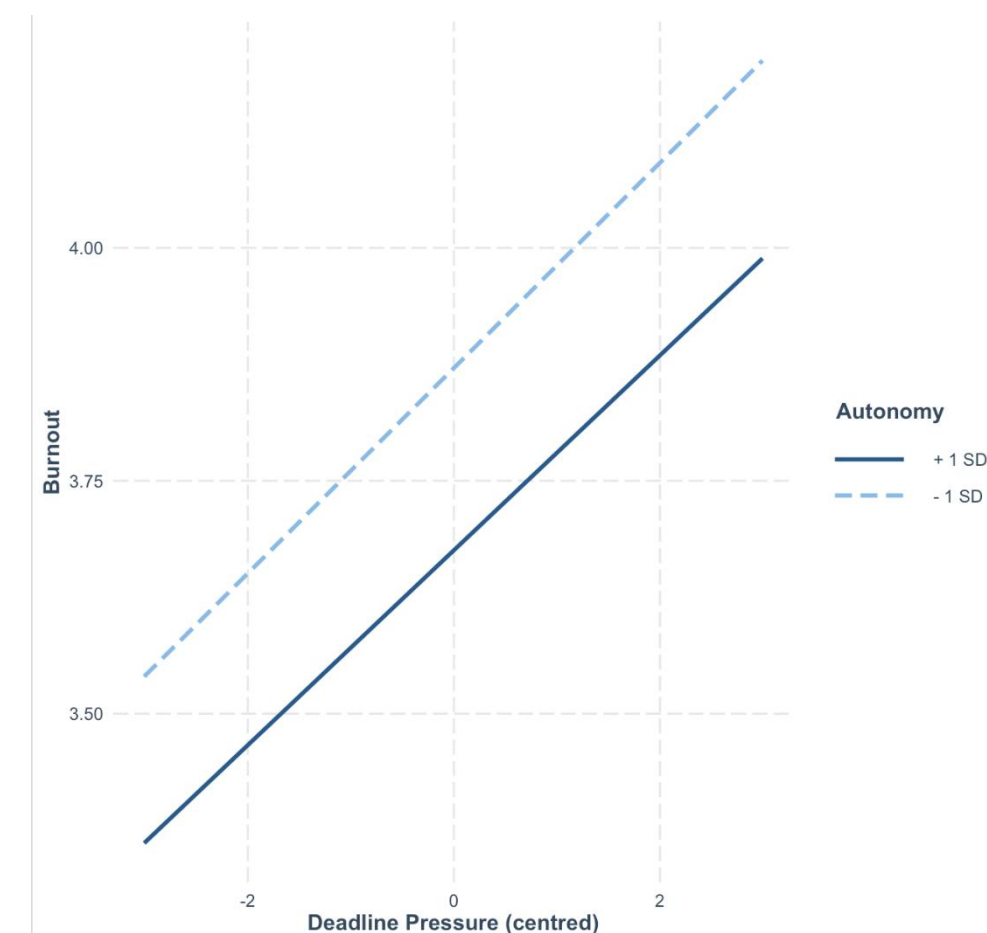
- **Interaction effect  $\beta_3$  :**  
Deadline pressure  $\times$  Autonomy  $\rightarrow$  Burnout  
Checks whether autonomy changes the strength of the deadline pressure–burnout relationship.

## Hypothesis 3

Autonomy moderates the relationship between deadline pressure and burnout, such that the positive relationship between deadline pressure and burnout is weaker when employee autonomy is high.



| Test                                      | Key result                                | Interpretation                                            |
|-------------------------------------------|-------------------------------------------|-----------------------------------------------------------|
| Interaction: Deadline pressure × Autonomy | $\beta = -0.002, p = .769$                | Not significant. H3 is not supported.                     |
| Autonomy main effect                      | $\beta = -0.062, p < .001$                | Autonomy directly reduces burnout.                        |
| $\Delta R^2$ test                         | $F = 0.086, p = .769$                     | Adding the interaction does not improve the model.        |
| Simple slopes                             | Low autonomy = .110; High autonomy = .105 | Slopes are almost identical, meaning no buffering effect. |



**Autonomy acts as a direct buffer, not a moderator** The interaction term was non-significant ( $\beta = -.002, p = .769, \Delta R^2 \approx .000$ ). Autonomy does not weaken the deadline–burnout relationship. However, autonomy has a significant negative main effect on burnout ( $\beta = -.062, p < .001$ ) — employees with higher autonomy report lower burnout overall, regardless of deadline pressure level.

Quantitative Data - Mediation and Moderation

AI TOOL RESEARCH

| Model               | AI tools $\beta$ | p     |
|---------------------|------------------|-------|
| Model 1 Baseline    | +.147            | <.001 |
| Model 2 Extended    | +.098            | <.001 |
| H2 Moderation model | +.147            | <.001 |
| H3 Moderation model | +.145            | <.001 |

Across all regression models, daily AI tool users consistently report higher burnout than non-users ( $\beta \approx +.10$  to  $+.15$ ,  $p < .001$ ). This pattern holds after controlling for deadline pressure, work hours, manager support, autonomy, work mode, and demographics, **suggesting AI tool use is independently associated with elevated burnout risk.**

However

So far, **only test the main effect of AI use** , they do not test whether AI use **changes how strongly demands translate into burnout.**  
A formal moderation test is needed to answer RQ4: Does daily AI tool use moderate the relationship between work demands and burnout — either amplifying it (more cognitive load) or buffering it (efficiency gains) fully.

## Hypothesis 4

AI tool use moderates the relationship between work demands and burnout, such that the positive relationship between deadline pressure/work hours and burnout is either stronger (H4a: cognitive overload) or weaker (H4b: efficiency gains) for daily AI tool users.

|     | Hypothesis                                       | Expected sign of $\beta_3$ |
|-----|--------------------------------------------------|----------------------------|
| H4a | AI amplifies demand–burnout (cognitive overload) | Positive $\beta_3$         |
| H4b | AI buffers demand–burnout (efficiency gains)     | Negative $\beta_3$         |

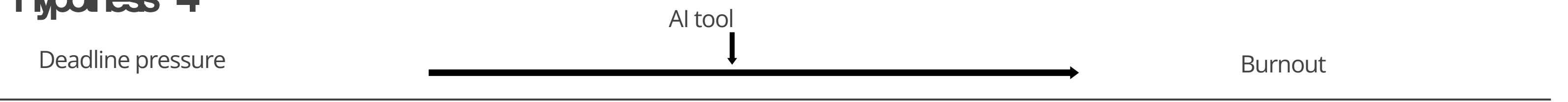
## 01. Research design

$$\text{Burnout} = \beta_0 + \beta_1 \text{DeadlinePressure} + \beta_2 \text{AI\_use} + \beta_3 (\text{DeadlinePressure} \times \text{AI\_use}) + \text{Controls} + \varepsilon$$

| Role                            | Variable                                                                        |
|---------------------------------|---------------------------------------------------------------------------------|
| <b>X / Independent variable</b> | Deadline pressure                                                               |
| <b>Y / Dependent variable</b>   | Burnout                                                                         |
| <b>Moderator / W</b>            | Daily AI tool use (0=No, 1=Yes)                                                 |
| <b>Controls</b>                 | Work hours per week, work mode, AI tools daily, age, tenure, hierarchical level |

- **Interaction effect  $\beta_3$  :**  
Deadline pressure  $\times$  AI tool use  $\rightarrow$  Burnout
- Checks whether being a daily AI user changes how strongly deadline pressure translates into burnout.

# Hypothesis 4



## AI tools

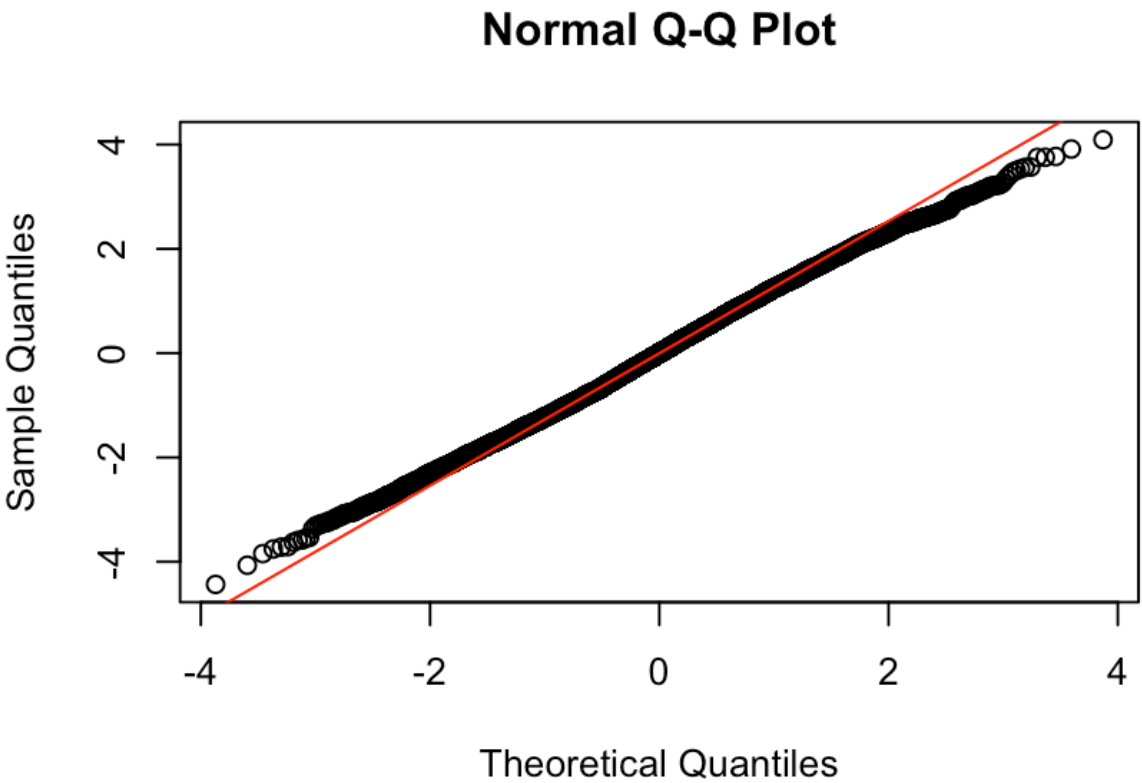
| Test                                      | Key result                     | Interpretation                                          |
|-------------------------------------------|--------------------------------|---------------------------------------------------------|
| Interaction: Deadline pressure × AI tools | $\beta = -.026, p = .223$      | Not significant. H4 not supported for deadline pathway. |
| AI tools main effect                      | $\beta = +.145, p < .001$      | AI users report higher burnout overall.                 |
| $\Delta R^2$ test                         | $F = 1.485, p = .223$          | Adding the interaction does not improve the model.      |
| Simple slopes                             | Non-AI = .127; AI users = .101 | Slopes differ slightly but not significantly.           |

| Test                               | Key result                     | Interpretation                                               |
|------------------------------------|--------------------------------|--------------------------------------------------------------|
| Interaction: Work hours × AI tools | $\beta = +.018, p = .335$      | Not significant. H4 not supported for hours pathway.         |
| $\Delta R^2$ test                  | $F = 0.928, p = .335$          | Adding the interaction does not improve the model.           |
| Simple slopes                      | Non-AI = .104; AI users = .110 | Slopes almost identical — no buffering or amplifying effect. |

AI tool use acts as a direct burnout risk factor, not a moderator. Neither interaction term was significant — Deadline × AI ( $\beta = -.026, p = .223$ ) nor WorkHours × AI ( $\beta = +.018, p = .335$ ). AI tool use does not change how strongly demands translate into burnout. However, AI daily users consistently report higher burnout overall ( $\beta = +.145, p < .001$ ), regardless of demand level. This suggests the issue is not that AI amplifies demand sensitivity, but that AI-associated roles carry persistently higher baseline burnout risk — pointing to workload and role design rather than AI itself as the intervention target.

# Quantitative Data-Diagnostics and Assumption Checks

| Check                               | Result                         | Threshold            | Interpretation                            |
|-------------------------------------|--------------------------------|----------------------|-------------------------------------------|
| Multicollinearity (VIF)             | All VIF 1.00–1.25              | < 5 (ideally < 3)    | No multicollinearity concern              |
| Influential observations (Cook's D) | 380 obs exceed threshold       | $4/(n-k-1) = .00043$ | Sensitivity check confirms results robust |
| Normality of residuals (Q-Q plot)   | Points closely follow diagonal | Visual inspection    | Approximately normal residuals            |
| Homoscedasticity (Scale-Location)   | Flat red line                  | No fan pattern       | Constant variance confirmed               |



The regression model meets all key assumptions , no multicollinearity, approximately normal residuals, and robust results confirmed by sensitivity analysis, supporting the validity and reliability of the reported findings.

# Quantitative Data research result

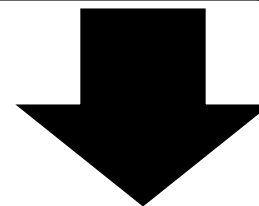
Overall quantitative finding

The quantitative analysis shows that **burnout is mainly driven by stress and workload-related demands**. Deadline pressure matters, but its effect appears to operate mainly **through stress** rather than directly. Manager support and autonomy reduce burnout directly, but they do **not** significantly buffer the effect of deadline pressure. AI tool use is associated with higher burnout as a main effect, but the AI moderation test still needs to be shown separately if RQ4 is included.

| Research question              | Finding                                                                                                                                                                      | Conclusion                                                                                                                                                                                                                                                                            |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RQ1 Direct effects             | Deadline pressure and work hours predict burnout in the baseline model. In the extended model, work hours remain significant, but deadline pressure becomes non-significant. | Workload matters directly; deadline pressure works mainly through stress.                                                                                                                                                                                                             |
| RQ2 Mediation                  | Deadline pressure → Stress → Burnout is significant; indirect effect $\beta = .116$ , $p < .001$ .                                                                           | Stress mediates the deadline pressure–burnout relationship. H1 supported.                                                                                                                                                                                                             |
| RQ3 Manager support moderation | Interaction $\beta = -.003$ , $p = .658$ . Manager support main effect $\beta = -.118$ , $p < .001$ .                                                                        | H2 not supported. Manager support reduces burnout directly but does not buffer deadline pressure.                                                                                                                                                                                     |
| RQ4 AI moderation              | AI use has a significant positive main effect on burnout. Interaction test still needed for moderation.                                                                      | H3,H4 not supported.<br>RQ4 is not supported as a moderation effect. Daily AI tool use does not significantly amplify or buffer the relationship between work demands and burnout. Neither the deadline pressure × AI interaction nor the work hours × AI interaction is significant. |

# Development research- Quantitative to Qualitative

| Quant finding                                    | What it tells us                                               | What remains unclear                                                                | Qualitative follow-up focus                                                                       |
|--------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Deadline pressure affects burnout through stress | The key mechanism is psychological strain.                     | What exactly creates stress from deadlines?                                         | Ask employees about deadline pressure, shifting priorities, recovery, and unclear expectations.   |
| Work hours directly predict burnout              | Workload intensity matters.                                    | Why are long hours happening?                                                       | Explore workload planning, meeting load, after-hours work, and staffing.                          |
| Manager support reduces burnout directly         | Support is useful, but not a buffer against deadline pressure. | What kind of support actually helps? Why is support not enough under high pressure? | Interview employees and managers about prioritization, delegation, feedback, and barrier removal. |
| Autonomy reduces burnout directly                | Autonomy helps, but does not neutralize deadline pressure.     | What type of autonomy is useful?                                                    | Explore task autonomy, scheduling flexibility, decision influence, and constraints.               |
| AI users report higher burnout                   | AI-related roles may carry higher baseline burnout risk.       | Is AI creating cognitive load, speed expectations, or role complexity?              | Ask AI users about learning burden, productivity pressure, monitoring, and task redesign.         |



Why do these patterns exist in daily work, and what practical changes would reduce burnout?  
 RQ5: How do employees and managers explain the mechanisms through which deadline pressure and work hours create stress and burnout, and how do managerial support, autonomy, and AI tool use shape these experiences in daily work?

# Qualitative research – RQ5

---

## Semi-structured Interview design

**Interview Format:** Semi-structured

**Scope:**

Employees and managers from different work modes, business areas, and burnout levels. The interviews will focus on how deadline pressure, work hours, stress, manager support, autonomy, and daily AI tool use are experienced in everyday work.

**Purpose of data collection:**

To explain the quantitative findings behind RQ1–RQ4. The survey shows that stress mediates the deadline pressure–burnout relationship, while manager support and autonomy reduce burnout directly but do not buffer deadline pressure. The interviews will explore why these patterns occur and what practical work-design changes could reduce burnout.

**Potential bias:**

Self-report bias, social desirability bias, recall bias, and selection bias. Employees may underreport stress or criticize management less openly. To reduce this, interviews should be anonymized, questions should be neutral, and participants should be selected across high- and low-burnout groups.

# Employee interviews

---

## Design logic — why interviews

- The survey shows where and how strong the effects are; interviews explain the why behind the regression, mediation and moderation results.
- Directly addresses our cross-sectional limitation
- Inductive, Gioia-inspired coding feeds our 1st-order → 2nd-order → aggregate-dimension structure.

## What we explore (mapped to constructs)

- What “deadline pressure” means day-to-day: real deadlines vs shifting priorities (the DEAD2 ambiguity), unexpected tasks.
- Sources of strain: meeting overload, unclear priorities, workload, cross-time-zone work, AI-tool adoption (friction vs relief).
- Resources in action: how supported / autonomous staff actually recover; a validity check on the excluded weak items.

## Sampling — data-driven contrasting cases

- Purposive sampling on the burnout factor scores from our CFA model — not a random draw.
- Two contrasting groups: high-burnout vs low-burnout (top / bottom tertile).
- Match both on deadline pressure & work hours, so both face high demand: isolating why some employees report lower burnout despite similar demands, even though the survey did not support support/autonomy as statistical buffers.
- Scope: 15–20 interviews, stratified across work mode and Business Area

## Why it matters

- Turns coefficients into mechanisms executives can act on, and surfaces resilience practices we can scale.
- Quant → Qual bridge: the strongest survey paths set the focus of the interview.

# Manager focus groups

---

## Design logic — why managers

- Manager support is employee-rated in the survey, so we only see the demand side. Managers are a distinct unit of analysis: do they have the time, tools and mandate to support staff and reduce demands?
- Two complementary modes: focus groups for shared, team-level norms; a diary study for daily within-person variation.

## Manager focus groups

- 3–5 groups, 5–6 managers each, across divisions and seniority
- Discuss: workload planning, delegation, meeting culture, feedback and barrier removal.
- Purpose: why deadline pressure is hard to reduce, what blocks delegation and autonomy, and practical, ownable improvement ideas.

## Manager diary study

- About 2 weeks of short daily entries from a manager subsample.
- Log: meeting load, admin / rhythm-of-business vs coaching, after-hours work, focus time, perceived pressure.

## Why it matters

- Targets interventions at the work system, not just individuals: workload planning, meeting hygiene, delegation mandates.
- Manager ownership of the solutions raises implementation success.
- Analysis: thematic analysis (focus groups) + descriptive pattern analysis (diary)

## VIVA Discussion

# Should Annual Surveys Be Replaced by Microsoft Viva Insights?

---

- Strengths vs Weaknesses
- Validity
- Reliability
- Ethical Considerations
- Implementation Challenges

# Should Annual Surveys Be Replaced by Microsoft Viva Insights?

## Strengths vs Weakness

---

### VIVA STRENGTHS

- Objective, continuous, high-frequency data
- No recall bias or social desirability bias
- Covers entire workforce automatically
- Reveals hidden behavioral patterns
- Example: Viva identified that large redundant meetings pushed engineers to work outside normal hours (Viva e-book)

### VIVA WEAKNESSES — What It Misses

- Motivation and psychological safety
- Stress, burnout, and emotional exhaustion
- Trust in leadership
- Perceived fairness and autonomy
- Emotional well-being
- Context behind behavior
- (the 'why' people act as they do)

# Should Annual Surveys Be Replaced by Microsoft Viva Insights?

## Validity Comparison

---

| Construct         | Survey measures                    | Viva measures                        | Gap                                         |
|-------------------|------------------------------------|--------------------------------------|---------------------------------------------|
| Burnout           | Emotional exhaustion (self-report) | After-hours work, low focus time     | Behavioral proxies not burnout itself       |
| Stress            | Felt tension, overwhelm            | Meeting overload, response time      | Observable correlates not subjective stress |
| Deadline pressure | Perceived time pressure            | Calendar density, task switching     | Indirect proxy only                         |
| Manager support   | Perceived support quality          | Manager 1:1 frequency, response time | Frequency is not equal to quality           |
| Autonomy          | Perceived task control             | Not measurable in trace data         | No equivalent metric                        |

# Should Annual Surveys Be Replaced by Microsoft Viva Insights?

## Reliability Comparison

---

### VIVA INSIGHTS — Very Reliable

- Automated — no human input error
- Continuous — same method every day
- No human bias in data collection
- Consistent measurement definition
- Timestamped behavioral log

### ANNUAL SURVEY — Less Reliable

- Mood and energy at time of answering
- Timing effects (busy vs quiet period)
- Item interpretation varies by person
- Survey fatigue reduces quality
- In our study:  $\alpha = .732-.804$
- (acceptable but not perfect)

# Should Annual Surveys Be Replaced by Microsoft Viva Insights?

## Ethical Considerations

---

|                                                                    |                                                                 |                                                                        |
|--------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Provenance</b>                                                  | <b>Purpose</b>                                                  | <b>Protection</b>                                                      |
| Survey: Employees know what they answer                            | Survey: Clear — annual HR review                                | Survey: Anonymous, aggregated responses                                |
| Viva: Passive collection — employees may not know what is captured | Viva: Risk of scope creep: wellbeing → performance surveillance | Viva: Individual-level behavioral data — higher re-identification risk |
| <b>Privacy</b>                                                     | <b>Preparation</b>                                              |                                                                        |
| Survey: Opt-in, voluntary participation                            | Survey: Employees briefed before survey                         |                                                                        |
| Viva: Passive — implicit rather than explicit consent              | Viva: Training needed on what Viva tracks and how data is used  |                                                                        |

# Should Annual Surveys Be Replaced by Microsoft Viva Insights?

## Implementation Challenges

---

### 01 GDPR Compliance & Data Governance

Behavioral trace data requires strict governance frameworks, data minimization policies, and legal basis for processing under GDPR.

### 03 Loss of Employee Voice

Surveys allow employees to express concerns, dissatisfaction, and ideas. Viva data is passive, employees cannot use it to communicate.

### 05 Need for Manager Training

Leaders need training to interpret behavioral data responsibly and avoid punitive use of insights about individual work patterns.

### 02 Cultural Resistance

Employees may perceive monitoring as surveillance "Are we being watched?" Trust must be established before adoption can succeed.

### 04 Over-Optimisation Risk

Managers may optimise for what is measurable rather than what matters. e.g. "cut meeting hours by 20%" without addressing why meetings happen.

### 06 Loss of Longitudinal Benchmarks

Replacing the survey loses years of trend data. Satisfaction and burnout benchmarks built over time cannot be reconstructed from trace data.

# Should Annual Surveys Be Replaced by Microsoft Viva Insights?

**Recommendation: Do NOT replace surveys. Combine them.**

---

## CONTINUOUS

### Viva Insights

Behavioral signals: meeting load, after-hours work, focus time, collaboration patterns

*Early warning system — triggers action when signals spike*

**Real-time**

Continuous

## ANNUAL + QUARTERLY

### Survey

Psychological experience: burnout, stress, autonomy, manager support — shorter & more focused annual + quarterly pulse

*Captures what people feel — irreplaceable for our constructs*

**Structured**

Annual + Quarterly

## TARGETED

### Qualitative Follow-ups

Interviews and focus groups to explain WHY the behavioral and survey patterns emerge

*Bridges quant findings with lived experience*

**Explanatory**

Targeted

# Sum up and actions

---

| Key insight                                                          | Recommended action                                                                                                    | Expected business value                                                             |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Stress is the main pathway from deadline pressure to burnout.        | Improve deadline planning, clarify priorities, and reduce last-minute task changes before pressure turns into stress. | Lower burnout risk and more sustainable performance.                                |
| Work hours remain a direct burnout driver.                           | Monitor excessive working hours, after-hours work, and workload allocation across teams.                              | Prevent overwork and reduce long-term absence or turnover risk.                     |
| Manager support reduces burnout directly.                            | Train managers to remove barriers, give clearer feedback, prioritize tasks, and support recovery.                     | Stronger employee resilience and better team performance.                           |
| Autonomy reduces burnout directly, but does not neutralize pressure. | Increase task autonomy and decision influence, but combine this with workload redesign.                               | Employees gain more control without being left alone with high demands.             |
| AI users report higher burnout overall.                              | Review AI-intensive roles for hidden workload, learning burden, speed expectations, and role complexity.              | Better AI implementation and lower risk of digital overload.                        |
| Support, autonomy, and AI do not buffer high demands.                | Do not rely only on individual resources or tools; reduce the underlying work demands.                                | Moves burnout prevention from “well-being support” to real work-system improvement. |

# Thank You

---



# Appendix

---

# Appendix 1: Review and quality check

```
> nrow(df) # expected: 9,230
[1] 9230
> ncol(df) # expected: 33 variables
[1] 33
> names(df) # confirm all column names
[1] "work_hours_per_week" "ai_tools_daily" "work_mode"
[4] "hierarchical_level" "age" "gender"
[7] "tenure" "Business_Area" "BURN1"
[10] "BURN2" "BURN3" "BURN4"
[13] "BURN5" "STR1" "STR2"
[16] "STR3" "STR4" "STR5"
[19] "DEAD1" "DEAD2" "DEAD3"
[22] "DEAD4" "DEAD5" "MSUP1"
[25] "MSUP2" "MSUP3" "MSUP4"
[28] "MSUP5" "AUT01" "AUT02"
[31] "AUT03" "AUT04" "AUT05"
```

```
> # — 1. Item-level descriptives for all 25 Likert items
> describe(df[, likert_items])[, c("n", "mean", "sd", "min", "max", "skew", "kurtosis")]
      n mean  sd min max skew kurtosis
BURN1 9230 3.99 2.02  1  7  0.01  -1.26
BURN2 9230 4.01 2.00  1  7 -0.01  -1.25
BURN3 9230 4.00 2.00  1  7  0.00  -1.24
BURN4 9230 4.00 2.01  1  7  0.00  -1.26
BURN5 9230 3.99 2.00  1  7  0.00  -1.26
STR1  9230 3.97 2.00  1  7  0.03  -1.25
STR2  9230 3.95 2.00  1  7  0.04  -1.25
STR3  9230 3.98 2.01  1  7  0.01  -1.26
STR4  9230 3.96 2.01  1  7  0.03  -1.27
STR5  9230 4.00 2.01  1  7  0.00  -1.26
DEAD1 9230 4.00 2.00  1  7  0.00  -1.26
DEAD2 9230 4.00 2.00  1  7  0.00  -1.24
DEAD3 9230 4.02 2.00  1  7 -0.01  -1.26
DEAD4 9230 4.01 2.00  1  7  0.01  -1.24
DEAD5 9230 3.99 2.01  1  7  0.01  -1.27
MSUP1 9230 4.01 2.00  1  7  0.00  -1.25
MSUP2 9230 3.99 1.99  1  7  0.01  -1.24
MSUP3 9230 3.99 1.99  1  7  0.01  -1.24
MSUP4 9230 3.99 2.01  1  7  0.01  -1.25
MSUP5 9230 4.00 1.99  1  7 -0.01  -1.25
AUT01 9230 3.98 2.00  1  7  0.00  -1.25
AUT02 9230 4.01 2.00  1  7  0.00  -1.24
AUT03 9230 4.01 2.00  1  7 -0.01  -1.25
AUT04 9230 4.01 2.01  1  7 -0.01  -1.27
AUT05 9230 3.99 2.00  1  7  0.01  -1.25
> # All item means expected near 4.0 (midpoint of 1–7 scale)
> # Skew near 0 confirms symmetric response distribution
>
```

## Appendix 2: Review and quality check -Code

```
> # — 2. Contextual variable descriptives —————
> describe(df[, c("work_hours_per_week", "age", "tenure", "hierarchical_level")])[,
+                                                     c("n", "mean", "sd", "min", "max")]
               n  mean   sd  min max
work_hours_per_week 9230 47.08 5.23 35.0  68
age                  9230 31.34 6.37 22.0  55
tenure               9230  2.47 2.44  0.1  15
hierarchical_level  9230  2.78 1.42  1.0   6
> # work_hours mean ≈ 47.1 hrs/week (range 35–68) – notable overwork signal
> # age mean ≈ 31.3 years; tenure mean ≈ 2.5 years (young, relatively new workforce)
>
> # — 3. Burnout by work mode – executive insight —————
> df %>% group_by(work_mode) %>%
+   summarise(n = n(),
+             Burnout_mean = round(mean(rowMeans(across(c(BURN1, BURN2, BURN3, BURN5))))), 3),
+             WorkHrs_mean = round(mean(work_hours_per_week), 1))
# A tibble: 3 × 4
  work_mode      n Burnout_mean WorkHrs_mean
  <fct>      <int>      <dbl>      <dbl>
1 On-site    2235         3.91         47.2
2 Hybrid     3686         4.00         47.1
3 Remote     3309         4.05          47
> # Expected burnout: Remote=4.05 > Hybrid=4.00 > On-site=3.91
```

# Appendix 3: CRONBACH'S ALPHA

---

```
> # — Alpha-if-item-deleted (identify noisy items) —————
> for (nm in names(cons)) {
+   cat("\n=====", nm, "=====\n")
+   print(psych::alpha(df[cons[[nm]]])$alpha.drop[, "raw_alpha", drop=FALSE])
+ }
```

```
==== Burnout ====
      raw_alpha
BURN1 0.7559264
BURN2 0.7588896
BURN3 0.7598930
BURN4 0.7933787
BURN5 0.7621106
```

```
==== Stress ====
      raw_alpha
STR1 0.7182602
STR2 0.7189549
STR3 0.7107342
STR4 0.7165516
STR5 0.7976995
```

```
==== Deadline ====
      raw_alpha
DEAD1 0.6525716
DEAD2 0.7697134
DEAD3 0.6640529
DEAD4 0.6640626
DEAD5 0.6638270
```

```
==== MgrSupp ====
      raw_alpha
MSUP1 0.6820863
MSUP2 0.6876103
MSUP3 0.7761140
MSUP4 0.6874429
MSUP5 0.6869900
```

```
==== Autonomy ====
      raw_alpha
AUT01 0.7908589
AUT02 0.6878599
AUT03 0.6913615
AUT04 0.6972045
AUT05 0.6922707
```

# Appendix 4: CRONBACH'S ALPHA Removed items (alpha if delete in appendix)

> alpha\_candidates

|    | Construct | Current_alpha | Item  | Alpha_if_deleted | Alpha_change |
|----|-----------|---------------|-------|------------------|--------------|
| 1  | Burnout   | 0.804         | BURN1 | 0.756            | -0.048       |
| 2  | Burnout   | 0.804         | BURN2 | 0.759            | -0.045       |
| 3  | Burnout   | 0.804         | BURN3 | 0.760            | -0.044       |
| 4  | Burnout   | 0.804         | BURN4 | 0.793            | -0.011       |
| 5  | Burnout   | 0.804         | BURN5 | 0.762            | -0.042       |
| 6  | Stress    | 0.776         | STR1  | 0.718            | -0.057       |
| 7  | Stress    | 0.776         | STR2  | 0.719            | -0.057       |
| 8  | Stress    | 0.776         | STR3  | 0.711            | -0.065       |
| 9  | Stress    | 0.776         | STR4  | 0.717            | -0.059       |
| 10 | Stress    | 0.776         | STR5  | 0.798            | 0.022        |
| 11 | Deadline  | 0.732         | DEAD1 | 0.653            | -0.080       |
| 12 | Deadline  | 0.732         | DEAD2 | 0.770            | 0.038        |
| 13 | Deadline  | 0.732         | DEAD3 | 0.664            | -0.068       |
| 14 | Deadline  | 0.732         | DEAD4 | 0.664            | -0.068       |
| 15 | Deadline  | 0.732         | DEAD5 | 0.664            | -0.068       |
| 16 | MgrSupp   | 0.750         | MSUP1 | 0.682            | -0.068       |
| 17 | MgrSupp   | 0.750         | MSUP2 | 0.688            | -0.063       |
| 18 | MgrSupp   | 0.750         | MSUP3 | 0.776            | 0.026        |
| 19 | MgrSupp   | 0.750         | MSUP4 | 0.687            | -0.063       |
| 20 | MgrSupp   | 0.750         | MSUP5 | 0.687            | -0.063       |
| 21 | Autonomy  | 0.758         | AUT01 | 0.791            | 0.033        |
| 22 | Autonomy  | 0.758         | AUT02 | 0.688            | -0.070       |
| 23 | Autonomy  | 0.758         | AUT03 | 0.691            | -0.067       |
| 24 | Autonomy  | 0.758         | AUT04 | 0.697            | -0.061       |
| 25 | Autonomy  | 0.758         | AUT05 | 0.692            | -0.066       |

>

# Appendix 5: CFA

---

```
> model_cfa <- '  
+   Burnout =~ BURN1 + BURN2 + BURN3 + BURN5  
+   Stress  =~ STR1  + STR2  + STR3  + STR4  
+   Deadline =~ DEAD1 + DEAD3 + DEAD4 + DEAD5  
+   MgrSupp =~ MSUP1 + MSUP2 + MSUP4 + MSUP5  
+   Autonomy =~ AUTO2 + AUTO3 + AUTO4 + AUTO5  
+ '  
> # BURN4, STR5, DEAD2, MSUP3, AUTO1 excluded – Step 04  
>  
> fit_cfa <- cfa(model_cfa, data=df,  
+               estimator="MLR",  # robust ML – handles mild non-normality  
+               missing="fiml",   # full-information ML (best practice)  
+               std.lv=TRUE)      # standardize latent variables  
>  
> # — Model fit indices (always use .robust versions with MLR) —————  
> fitMeasures(fit_cfa, c("cfi.robust", "tli.robust",  
+                       "rmsea.robust", "rmsea.ci.lower.robust", "rmsea.ci.upper.robust",  
+                       "srmr"))
```

| cfi.robust | tli.robust | rmsea.robust | rmsea.ci.lower.robust | rmsea.ci.upper.robust | srmr  |
|------------|------------|--------------|-----------------------|-----------------------|-------|
| 1.000      | 1.000      | 0.000        | 0.000                 | 0.005                 | 0.007 |

# Appendix 6: CFA - ML

```
> # — Factor loadings (standardized)
> summary(fit_cfa, standardized=TRUE, fit.measures=TRUE)
lavaan 0.6-21 ended normally after 22 iterations
```

|                            |        |
|----------------------------|--------|
| Estimator                  | ML     |
| Optimization method        | NLMINB |
| Number of model parameters | 70     |
| Number of observations     | 9230   |
| Number of missing patterns | 1      |

Model Test User Model:

|                                         | Standard | Scaled  |
|-----------------------------------------|----------|---------|
| Test Statistic                          | 159.066  | 157.310 |
| Degrees of freedom                      | 160      | 160     |
| P-value (Chi-square)                    | 0.506    | 0.545   |
| Scaling correction factor               |          | 1.011   |
| Yuan-Bentler correction (Mplus variant) |          |         |

Model Test Baseline Model:

|                           |           |           |
|---------------------------|-----------|-----------|
| Test statistic            | 54230.359 | 52829.240 |
| Degrees of freedom        | 190       | 190       |
| P-value                   | 0.000     | 0.000     |
| Scaling correction factor |           | 1.027     |

User Model versus Baseline Model:

|                             |       |       |
|-----------------------------|-------|-------|
| Comparative Fit Index (CFI) | 1.000 | 1.000 |
| Tucker-Lewis Index (TLI)    | 1.000 | 1.000 |

# Appendix 7: CFA

|                                    |       |
|------------------------------------|-------|
| Robust Comparative Fit Index (CFI) | 1.000 |
| Robust Tucker-Lewis Index (TLI)    | 1.000 |

## Loglikelihood and Information Criteria:

|                                                     |             |             |
|-----------------------------------------------------|-------------|-------------|
| Loglikelihood user model (H0)                       | -363054.083 | -363054.083 |
| Scaling correction factor<br>for the MLR correction | 0.868       |             |
| Loglikelihood unrestricted model (H1)               | -362974.550 | -362974.550 |
| Scaling correction factor<br>for the MLR correction | 0.967       |             |
| Akaike (AIC)                                        | 726248.165  | 726248.165  |
| Bayesian (BIC)                                      | 726747.280  | 726747.280  |
| Sample-size adjusted Bayesian (SABIC)               | 726524.832  | 726524.832  |

## Root Mean Square Error of Approximation:

|                                               |       |       |
|-----------------------------------------------|-------|-------|
| RMSEA                                         | 0.000 | 0.000 |
| 90 Percent confidence interval - lower        | 0.000 | 0.000 |
| 90 Percent confidence interval - upper        | 0.005 | 0.005 |
| P-value H <sub>0</sub> : RMSEA ≤ 0.050        | 1.000 | 1.000 |
| P-value H <sub>0</sub> : RMSEA ≥ 0.080        | 0.000 | 0.000 |
| Robust RMSEA                                  | 0.000 |       |
| 90 Percent confidence interval - lower        | 0.000 |       |
| 90 Percent confidence interval - upper        | 0.005 |       |
| P-value H <sub>0</sub> : Robust RMSEA ≤ 0.050 | 1.000 |       |
| P-value H <sub>0</sub> : Robust RMSEA ≥ 0.080 | 0.000 |       |

# Appendix 8: CFA

Standardized Root Mean Square Residual:

SRMR 0.007 0.007

Parameter Estimates:

Standard errors  
Information bread  
Observed information based on

Sandwich  
Observed  
Hessian

Latent Variables:

|             | Estimate | Std.Err | z-value | P(> z ) | Std.lv | Std.all |
|-------------|----------|---------|---------|---------|--------|---------|
| Burnout =~  |          |         |         |         |        |         |
| BURN1       | 1.428    | 0.017   | 84.547  | 0.000   | 1.428  | 0.708   |
| BURN2       | 1.402    | 0.017   | 83.910  | 0.000   | 1.402  | 0.702   |
| BURN3       | 1.377    | 0.017   | 80.837  | 0.000   | 1.377  | 0.690   |
| BURN5       | 1.399    | 0.017   | 83.176  | 0.000   | 1.399  | 0.699   |
| Stress =~   |          |         |         |         |        |         |
| STR1        | 1.394    | 0.017   | 83.165  | 0.000   | 1.394  | 0.696   |
| STR2        | 1.399    | 0.017   | 83.324  | 0.000   | 1.399  | 0.699   |
| STR3        | 1.450    | 0.016   | 88.814  | 0.000   | 1.450  | 0.723   |
| STR4        | 1.409    | 0.017   | 84.121  | 0.000   | 1.409  | 0.700   |
| Deadline =~ |          |         |         |         |        |         |
| DEAD1       | 1.419    | 0.018   | 78.085  | 0.000   | 1.419  | 0.710   |
| DEAD3       | 1.323    | 0.019   | 70.769  | 0.000   | 1.323  | 0.661   |
| DEAD4       | 1.331    | 0.019   | 71.618  | 0.000   | 1.331  | 0.667   |
| DEAD5       | 1.331    | 0.019   | 70.498  | 0.000   | 1.331  | 0.661   |
| MgrSupp =~  |          |         |         |         |        |         |
| MSUP1       | 1.395    | 0.018   | 76.626  | 0.000   | 1.395  | 0.696   |
| MSUP2       | 1.351    | 0.018   | 73.025  | 0.000   | 1.351  | 0.678   |
| MSUP4       | 1.359    | 0.018   | 73.587  | 0.000   | 1.359  | 0.678   |
| MSUP5       | 1.345    | 0.019   | 72.654  | 0.000   | 1.345  | 0.675   |

Autonomy =~

|       |       |       |        |       |       |       |
|-------|-------|-------|--------|-------|-------|-------|
| AUT02 | 1.424 | 0.018 | 80.562 | 0.000 | 1.424 | 0.713 |
| AUT03 | 1.407 | 0.018 | 79.261 | 0.000 | 1.407 | 0.703 |
| AUT04 | 1.368 | 0.018 | 75.615 | 0.000 | 1.368 | 0.680 |
| AUT05 | 1.383 | 0.018 | 77.259 | 0.000 | 1.383 | 0.692 |

Covariances:

|             | Estimate | Std.Err | z-value | P(> z ) | Std.lv | Std.all |
|-------------|----------|---------|---------|---------|--------|---------|
| Burnout ~~  |          |         |         |         |        |         |
| Stress      | 0.767    | 0.008   | 101.681 | 0.000   | 0.767  | 0.767   |
| Deadline    | 0.144    | 0.013   | 10.871  | 0.000   | 0.144  | 0.144   |
| MgrSupp     | -0.150   | 0.013   | -11.555 | 0.000   | -0.150 | -0.150  |
| Autonomy    | -0.063   | 0.013   | -4.737  | 0.000   | -0.063 | -0.063  |
| Stress ~~   |          |         |         |         |        |         |
| Deadline    | 0.278    | 0.013   | 21.997  | 0.000   | 0.278  | 0.278   |
| MgrSupp     | -0.103   | 0.013   | -7.900  | 0.000   | -0.103 | -0.103  |
| Autonomy    | -0.033   | 0.013   | -2.562  | 0.010   | -0.033 | -0.033  |
| Deadline ~~ |          |         |         |         |        |         |
| MgrSupp     | 0.009    | 0.014   | 0.660   | 0.509   | 0.009  | 0.009   |
| Autonomy    | -0.022   | 0.013   | -1.678  | 0.093   | -0.022 | -0.022  |
| MgrSupp ~~  |          |         |         |         |        |         |
| Autonomy    | -0.001   | 0.013   | -0.083  | 0.934   | -0.001 | -0.001  |

Intercepts:

|        | Estimate | Std.Err | z-value | P(> z ) | Std.lv | Std.all |
|--------|----------|---------|---------|---------|--------|---------|
| .BURN1 | 3.987    | 0.021   | 190.021 | 0.000   | 3.987  | 1.978   |
| .BURN2 | 4.008    | 0.021   | 192.867 | 0.000   | 4.008  | 2.008   |
| .BURN3 | 4.001    | 0.021   | 192.540 | 0.000   | 4.001  | 2.004   |
| .BURN5 | 3.988    | 0.021   | 191.512 | 0.000   | 3.988  | 1.993   |
| .STR1  | 3.972    | 0.021   | 190.605 | 0.000   | 3.972  | 1.984   |
| .STR2  | 3.949    | 0.021   | 189.641 | 0.000   | 3.949  | 1.974   |
| .STR3  | 3.982    | 0.021   | 190.771 | 0.000   | 3.982  | 1.986   |

# Appendix 9: CFA

|        |       |       |         |       |       |       |
|--------|-------|-------|---------|-------|-------|-------|
| .STR4  | 3.963 | 0.021 | 189.173 | 0.000 | 3.963 | 1.969 |
| .DEAD1 | 4.004 | 0.021 | 192.420 | 0.000 | 4.004 | 2.003 |
| .DEAD3 | 4.022 | 0.021 | 193.120 | 0.000 | 4.022 | 2.010 |
| .DEAD4 | 4.009 | 0.021 | 192.917 | 0.000 | 4.009 | 2.008 |
| .DEAD5 | 3.986 | 0.021 | 190.282 | 0.000 | 3.986 | 1.981 |
| .MSUP1 | 4.012 | 0.021 | 192.282 | 0.000 | 4.012 | 2.001 |
| .MSUP2 | 3.988 | 0.021 | 192.192 | 0.000 | 3.988 | 2.000 |
| .MSUP4 | 3.988 | 0.021 | 191.052 | 0.000 | 3.988 | 1.989 |
| .MSUP5 | 3.999 | 0.021 | 192.684 | 0.000 | 3.999 | 2.006 |
| .AUT02 | 4.006 | 0.021 | 192.610 | 0.000 | 4.006 | 2.005 |
| .AUT03 | 4.010 | 0.021 | 192.559 | 0.000 | 4.010 | 2.004 |
| .AUT04 | 4.007 | 0.021 | 191.507 | 0.000 | 4.007 | 1.993 |
| .AUT05 | 3.987 | 0.021 | 191.713 | 0.000 | 3.987 | 1.995 |

|          |       |       |        |       |       |       |
|----------|-------|-------|--------|-------|-------|-------|
| .AUT02   | 1.965 | 0.041 | 48.403 | 0.000 | 1.965 | 0.492 |
| .AUT03   | 2.023 | 0.041 | 49.583 | 0.000 | 2.023 | 0.505 |
| .AUT04   | 2.170 | 0.042 | 52.063 | 0.000 | 2.170 | 0.537 |
| .AUT05   | 2.078 | 0.041 | 51.290 | 0.000 | 2.078 | 0.521 |
| Burnout  | 1.000 |       |        |       | 1.000 | 1.000 |
| Stress   | 1.000 |       |        |       | 1.000 | 1.000 |
| Deadline | 1.000 |       |        |       | 1.000 | 1.000 |
| MgrSupp  | 1.000 |       |        |       | 1.000 | 1.000 |
| Autonomy | 1.000 |       |        |       | 1.000 | 1.000 |

Variances:

|        | Estimate | Std.Err | z-value | P(> z ) | Std.lv | Std.all |
|--------|----------|---------|---------|---------|--------|---------|
| .BURN1 | 2.025    | 0.038   | 52.991  | 0.000   | 2.025  | 0.498   |
| .BURN2 | 2.020    | 0.037   | 55.034  | 0.000   | 2.020  | 0.507   |
| .BURN3 | 2.090    | 0.037   | 56.293  | 0.000   | 2.090  | 0.524   |
| .BURN5 | 2.045    | 0.037   | 55.169  | 0.000   | 2.045  | 0.511   |
| .STR1  | 2.067    | 0.036   | 57.080  | 0.000   | 2.067  | 0.516   |
| .STR2  | 2.044    | 0.037   | 55.747  | 0.000   | 2.044  | 0.511   |
| .STR3  | 1.921    | 0.037   | 52.530  | 0.000   | 1.921  | 0.478   |
| .STR4  | 2.064    | 0.037   | 55.442  | 0.000   | 2.064  | 0.510   |
| .DEAD1 | 1.982    | 0.042   | 47.034  | 0.000   | 1.982  | 0.496   |
| .DEAD3 | 2.253    | 0.042   | 53.826  | 0.000   | 2.253  | 0.563   |
| .DEAD4 | 2.214    | 0.042   | 52.853  | 0.000   | 2.214  | 0.555   |
| .DEAD5 | 2.278    | 0.043   | 53.576  | 0.000   | 2.278  | 0.563   |
| .MSUP1 | 2.073    | 0.043   | 48.735  | 0.000   | 2.073  | 0.516   |
| .MSUP2 | 2.150    | 0.042   | 51.583  | 0.000   | 2.150  | 0.541   |
| .MSUP4 | 2.175    | 0.042   | 51.671  | 0.000   | 2.175  | 0.541   |
| .MSUP5 | 2.166    | 0.042   | 51.805  | 0.000   | 2.166  | 0.545   |

## Appendix 10: Post-CFA Reliability

---

```
> library(semTools)
>
> compRelSEM(fit_cfa, obs.var=TRUE, tau.eq=FALSE)
$Burnout
```

Composite `Burnout` is composed of observed variables:

BURN1, BURN2, BURN3, BURN5

True-score variance is represented by common factor(s):

Burnout

Total variance of composite `Burnout` determined from the unrestricted model.

The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):

[1] 0.793

```
$Stress
```

Composite `Stress` is composed of observed variables:

STR1, STR2, STR3, STR4

True-score variance is represented by common factor(s):

Stress

Total variance of composite `Stress` determined from the unrestricted model.

The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):

[1] 0.798

# Appendix 1: Post-CFA Reliability

---

\$Deadline

Composite `Deadline` is composed of observed variables:

DEAD1, DEAD3, DEAD4, DEAD5

True-score variance is represented by common factor(s):

Deadline

Total variance of composite `Deadline` determined from the unrestricted model.

The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):

[1] 0.77

\$MgrSupp

Composite `MgrSupp` is composed of observed variables:

MSUP1, MSUP2, MSUP4, MSUP5

True-score variance is represented by common factor(s):

MgrSupp

Total variance of composite `MgrSupp` determined from the unrestricted model.

The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):

[1] 0.776

\$Autonomy

Composite `Autonomy` is composed of observed variables:

AUT02, AUT03, AUT04, AUT05

True-score variance is represented by common factor(s):

Autonomy

Total variance of composite `Autonomy` determined from the unrestricted model.

The proportion attributable to "true" scores is its model-based estimate of reliability ("omega"):

[1] 0.791

# Appendix 12: Construct Scores

---

```
> df <- df %>% mutate(  
+   Burnout_avg = rowMeans(select(., BURN1, BURN2, BURN3, BURN5),      na.rm=TRUE),  
+   Stress_avg  = rowMeans(select(., STR1, STR2, STR3, STR4),          na.rm=TRUE),  
+   Deadline_avg = rowMeans(select(., DEAD1, DEAD3, DEAD4, DEAD5),      na.rm=TRUE),  
+   MgrSupp_avg = rowMeans(select(., MSUP1, MSUP2, MSUP4, MSUP5),        na.rm=TRUE),  
+   Autonomy_avg = rowMeans(select(., AUT02, AUT03, AUT04, AUT05),        na.rm=TRUE)  
+ )  
>  
> # — Confirm scores are on 1–7 scale  
> describe(df[, c("Burnout_avg", "Stress_avg", "Deadline_avg",  
+                 "MgrSupp_avg", "Autonomy_avg")][, c("mean", "sd", "min", "max")])  
      mean    sd min max  
Burnout_avg 4.00 1.57  1  7  
Stress_avg  3.97 1.58  1  7  
Deadline_avg 4.01 1.54  1  7  
MgrSupp_avg  4.00 1.55  1  7  
Autonomy_avg 4.00 1.57  1  7  
> # Expected: all means ≈ 4.0; all ranges 1.0–7.0  
> # Burnout_avg mean ≈ 3.996, sd ≈ 1.573  
> # Stress_avg mean ≈ 3.966, sd ≈ 1.582  
> # Deadline_avg mean ≈ 4.005, sd ≈ 1.540  
> # MgrSupp_avg mean ≈ 3.997, sd ≈ 1.546  
> # Autonomy_avg mean ≈ 4.002, sd ≈ 1.569  
>
```

# Appendix 13: Correlation Matrix

```
> cor_vars <- df[, c("Burnout_avg", "Stress_avg", "Deadline_avg",
+                    "MgrSupp_avg", "Autonomy_avg", "work_hours_per_week")]
>
> cor_out <- rcorr(as.matrix(cor_vars))
> round(cor_out$r, 3) # correlation coefficients
```

|                     | Burnout_avg | Stress_avg | Deadline_avg | MgrSupp_avg | Autonomy_avg | work_hours_per_week |
|---------------------|-------------|------------|--------------|-------------|--------------|---------------------|
| Burnout_avg         | 1.000       | 0.610      | 0.113        | -0.118      | -0.050       | 0.399               |
| Stress_avg          | 0.610       | 1.000      | 0.218        | -0.082      | -0.026       | 0.333               |
| Deadline_avg        | 0.113       | 0.218      | 1.000        | 0.007       | -0.018       | 0.018               |
| MgrSupp_avg         | -0.118      | -0.082     | 0.007        | 1.000       | 0.000        | -0.008              |
| Autonomy_avg        | -0.050      | -0.026     | -0.018       | 0.000       | 1.000        | 0.013               |
| work_hours_per_week | 0.399       | 0.333      | 0.018        | -0.008      | 0.013        | 1.000               |

```
> round(cor_out$p, 3) # p-values
```

|                     | Burnout_avg | Stress_avg | Deadline_avg | MgrSupp_avg | Autonomy_avg | work_hours_per_week |
|---------------------|-------------|------------|--------------|-------------|--------------|---------------------|
| Burnout_avg         | NA          | 0.000      | 0.000        | 0.000       | 0.000        | 0.000               |
| Stress_avg          | 0           | NA         | 0.000        | 0.000       | 0.012        | 0.000               |
| Deadline_avg        | 0           | 0.000      | NA           | 0.484       | 0.087        | 0.080               |
| MgrSupp_avg         | 0           | 0.000      | 0.484        | NA          | 0.990        | 0.462               |
| Autonomy_avg        | 0           | 0.012      | 0.087        | 0.990       | NA           | 0.225               |
| work_hours_per_week | 0           | 0.000      | 0.080        | 0.462       | 0.225        | NA                  |

```
>
```

# Appendix 14: Baseline Regression—Model 1

---

```
> # — Model 1: Demands + controls only —————
> m1 <- lm(Burnout_avg ~ Deadline_avg + work_hours_per_week +
+         age + tenure + hierarchical_level + work_mode + ai_tools_daily,
+         data = df)
>
> summary(m1)           # coefficients, R², F-test
```

Call:

```
lm(formula = Burnout_avg ~ Deadline_avg + work_hours_per_week +
    age + tenure + hierarchical_level + work_mode + ai_tools_daily,
    data = df)
```

Residuals:

|  | Min     | 1Q      | Median | 3Q     | Max    |
|--|---------|---------|--------|--------|--------|
|  | -4.3926 | -1.0807 | 0.0077 | 1.0690 | 3.8883 |

Coefficients:

|                     | Estimate   | Std. Error | t value | Pr(> t ) |     |
|---------------------|------------|------------|---------|----------|-----|
| (Intercept)         | -2.1432676 | 0.1606579  | -13.341 | < 2e-16  | *** |
| Deadline_avg        | 0.1084578  | 0.0095882  | 11.312  | < 2e-16  | *** |
| work_hours_per_week | 0.1088241  | 0.0029399  | 37.016  | < 2e-16  | *** |
| age                 | -0.0002093 | 0.0023183  | -0.090  | 0.92806  |     |
| tenure              | -0.0032335 | 0.0060568  | -0.534  | 0.59345  |     |
| hierarchical_level  | 0.1396598  | 0.0108492  | 12.873  | < 2e-16  | *** |
| work_modeHybrid     | 0.1060911  | 0.0380224  | 2.790   | 0.00528  | **  |
| work_modeRemote     | 0.1607337  | 0.0388286  | 4.140   | 3.51e-05 | *** |
| ai_tools_dailyYes   | 0.1469158  | 0.0331358  | 4.434   | 9.37e-06 | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.418 on 9221 degrees of freedom

Multiple R-squared: 0.1886, Adjusted R-squared: 0.1879

F-statistic: 268 on 8 and 9221 DF, p-value: < 2.2e-16

# Appendix 15: Baseline Regression—Model 1

```
> lm.beta(m1)           # standardized  $\beta$  for effect size comparison
```

Call:  
lm(formula = Burnout\_avg ~ Deadline\_avg + work\_hours\_per\_week +  
age + tenure + hierarchical\_level + work\_mode + ai\_tools\_daily,  
data = df)

Standardized Coefficients::

|                 |                 |                     |               |               |                    |
|-----------------|-----------------|---------------------|---------------|---------------|--------------------|
| (Intercept)     | Deadline_avg    | work_hours_per_week | age           | tenure        | hierarchical_level |
| NA              | 0.1061490503    | 0.3620689255        | -0.0008472458 | -0.0050086964 | 0.1259118878       |
| work_modeHybrid | work_modeRemote | ai_tools_dailyYes   |               |               |                    |
| 0.0330267291    | 0.0489947984    | 0.0416084119        |               |               |                    |

```
> confint(m1)           # 95% confidence intervals
```

|                     |              |              |
|---------------------|--------------|--------------|
|                     | 2.5 %        | 97.5 %       |
| (Intercept)         | -2.458192575 | -1.828342556 |
| Deadline_avg        | 0.089662907  | 0.127252707  |
| work_hours_per_week | 0.103061167  | 0.114586974  |
| age                 | -0.004753643 | 0.004334989  |
| tenure              | -0.015106273 | 0.008639195  |
| hierarchical_level  | 0.118393030  | 0.160926633  |
| work_modeHybrid     | 0.031558770  | 0.180623479  |
| work_modeRemote     | 0.084621093  | 0.236846219  |
| ai_tools_dailyYes   | 0.081962289  | 0.211869365  |

```
> vif(m1)               #  All VIF < 5 (ideally < 3)
```

|                     |          |    |                 |
|---------------------|----------|----|-----------------|
|                     | GVIF     | Df | GVIF^(1/(2*Df)) |
| Deadline_avg        | 1.000791 | 1  | 1.000395        |
| work_hours_per_week | 1.087347 | 1  | 1.042759        |
| age                 | 1.000595 | 1  | 1.000297        |
| tenure              | 1.000340 | 1  | 1.000170        |
| hierarchical_level  | 1.087296 | 1  | 1.042735        |
| work_mode           | 1.001445 | 2  | 1.000361        |
| ai_tools_daily      | 1.000884 | 1  | 1.000442        |

# Appendix 16: Extended Regression—Model 2

```
> # — Model 2: Adds stress, manager support, autonomy —————
> m2 <- lm(Burnout_avg ~ Deadline_avg + work_hours_per_week +
+         Stress_avg + MgrSupp_avg + Autonomy_avg +
+         age + tenure + hierarchical_level + work_mode + ai_tools_daily,
+         data = df)
>
> summary(m2)
```

Call:
lm(formula = Burnout\_avg ~ Deadline\_avg + work\_hours\_per\_week +
 Stress\_avg + MgrSupp\_avg + Autonomy\_avg + age + tenure +
 hierarchical\_level + work\_mode + ai\_tools\_daily, data = df)

Residuals:
 Min 1Q Median 3Q Max
-4.4344 -0.8590 -0.0011 0.8519 4.0917

Coefficients:

|                     | Estimate  | Std. Error | t value | Pr(> t ) |     |
|---------------------|-----------|------------|---------|----------|-----|
| (Intercept)         | -0.909809 | 0.142513   | -6.384  | 1.81e-10 | *** |
| Deadline_avg        | -0.005508 | 0.008303   | -0.663  | 0.507137 |     |
| work_hours_per_week | 0.065279  | 0.002583   | 25.268  | < 2e-16  | *** |
| Stress_avg          | 0.519410  | 0.008773   | 59.206  | < 2e-16  | *** |
| MgrSupp_avg         | -0.074615 | 0.008072   | -9.243  | < 2e-16  | *** |
| Autonomy_avg        | -0.043322 | 0.008114   | -5.339  | 9.56e-08 | *** |
| age                 | 0.001092  | 0.001953   | 0.559   | 0.576188 |     |
| tenure              | -0.001500 | 0.005102   | -0.294  | 0.768749 |     |
| hierarchical_level  | 0.032871  | 0.009316   | 3.528   | 0.000420 | *** |
| work_modeHybrid     | 0.079639  | 0.032033   | 2.486   | 0.012929 | *   |
| work_modeRemote     | 0.113868  | 0.033254   | 3.424   | 0.000619 | *** |
| ai_tools_dailyYes   | 0.097821  | 0.027923   | 3.503   | 0.000462 | *** |

---
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.194 on 9218 degrees of freedom
Multiple R-squared: 0.4245, Adjusted R-squared: 0.4238
F-statistic: 618.2 on 11 and 9218 DF, p-value: < 2.2e-16

# Appendix 17: Extended Regression—Model 2

```
> lm.beta(m2)
```

```
Call:
lm(formula = Burnout_avg ~ Deadline_avg + work_hours_per_week +
    Stress_avg + MgrSupp_avg + Autonomy_avg + age + tenure +
    hierarchical_level + work_mode + ai_tools_daily, data = df)
```

```
Standardized Coefficients:
      (Intercept)      Deadline_avg work_hours_per_week      Stress_avg
              NA      -0.005390488           0.217188716      0.522228151
              age           tenure hierarchical_level work_modeHybrid
      0.004418273      -0.002323676           0.029635595      0.024792183
              MgrSupp_avg      Autonomy_avg
      -0.073341317      -0.043208488
      work_modeRemote ai_tools_dailyYes
      0.034709276      0.027704250
```

```
> confint(m2)
```

```
                2.5 %      97.5 %
(Intercept) -1.189165423 -0.630453117
Deadline_avg -0.021783783  0.010768320
work_hours_per_week 0.060214436  0.070342806
Stress_avg      0.502213027  0.536606933
MgrSupp_avg     -0.090438329 -0.058791547
Autonomy_avg    -0.059227040 -0.027416789
age             -0.002736428  0.004919652
tenure          -0.011501448  0.008501187
hierarchical_level 0.014609315  0.051133520
work_modeHybrid  0.016848412  0.142430489
work_modeRemote  0.048682151  0.179054224
ai_tools_dailyYes 0.043086059  0.152556725
```

```
> vif(m2) # 🔴 Check VIF – Stress and Burnout are correlated (.61), watch for inflation
```

```
          GVIF Df GVIF^(1/(2*Df))
Deadline_avg 1.057806 1      1.028497
work_hours_per_week 1.183448 1      1.087864
Stress_avg    1.246236 1      1.116350
MgrSupp_avg   1.008422 1      1.004202
Autonomy_avg  1.049046 1      1.024229
age           1.000734 1      1.000367
tenure        1.000472 1      1.000236
hierarchical_level 1.130029 1      1.063028
work_mode     1.050726 2      1.012447
ai_tools_daily 1.001747 1      1.000873
```

```
>
```

# Appendix 18: Extended Regression—Model 2

```
> # —  $\Delta R^2$  test: Model 2 vs Model 1
> anova(m1, m2)
Analysis of Variance Table

Model 1: Burnout_avg ~ Deadline_avg + work_hours_per_week + age + tenure +
  hierarchical_level + work_mode + ai_tools_daily
Model 2: Burnout_avg ~ Deadline_avg + work_hours_per_week + Stress_avg +
  MgrSupp_avg + Autonomy_avg + age + tenure + hierarchical_level +
  work_mode + ai_tools_daily
  Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1    9221 18536
2    9218 13147  3    5388.9 1259.5 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

|   | lhs         | op | rhs          | label    | est    | se    | z      | pvalue | ci.lower | ci.upper | std.lv | std.all | std.nox |
|---|-------------|----|--------------|----------|--------|-------|--------|--------|----------|----------|--------|---------|---------|
| 1 | Stress_avg  | ~  | Deadline_avg | a        | 0.219  | 0.010 | 22.912 | 0.000  | 0.201    | 0.238    | 0.219  | 0.213   | 0.138   |
| 2 | Burnout_avg | ~  | Stress_avg   | b        | 0.530  | 0.008 | 62.559 | 0.000  | 0.513    | 0.547    | 0.530  | 0.533   | 0.533   |
| 3 | Burnout_avg | ~  | Deadline_avg | cprime   | -0.007 | 0.008 | -0.892 | 0.372  | -0.024   | 0.009    | -0.007 | -0.007  | -0.005  |
| 4 | indirect    | := | a*b          | indirect | 0.116  | 0.005 | 21.385 | 0.000  | 0.106    | 0.127    | 0.116  | 0.114   | 0.074   |
| 5 | direct      | := | cprime       | direct   | -0.007 | 0.008 | -0.892 | 0.372  | -0.024   | 0.009    | -0.007 | -0.007  | -0.005  |
| 6 | total       | := | cprime+a*b   | total    | 0.109  | 0.010 | 11.397 | 0.000  | 0.090    | 0.127    | 0.109  | 0.106   | 0.069   |

# Appendix 19: Moderation 1—Manager Support (H2)

Call:

```
lm(formula = Burnout_avg ~ Dead_c * Msup_c + WHrs_c + age + tenure +
    hierarchical_level + work_mode + ai_tools_daily, data = df)
```

Residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -4.1901 | -1.0563 | 0.0106 | 1.0476 | 3.9158 |

Coefficients:

|                    | Estimate   | Std. Error | t value | Pr(> t )     |
|--------------------|------------|------------|---------|--------------|
| (Intercept)        | 3.4180608  | 0.0881812  | 38.762  | < 2e-16 ***  |
| Dead_c             | 0.1093253  | 0.0095099  | 11.496  | < 2e-16 ***  |
| Msup_c             | -0.1176427 | 0.0094678  | -12.426 | < 2e-16 ***  |
| WHrs_c             | 0.1085183  | 0.0029160  | 37.215  | < 2e-16 ***  |
| age                | -0.0002329 | 0.0023003  | -0.101  | 0.91934      |
| tenure             | -0.0036496 | 0.0060074  | -0.608  | 0.54352      |
| hierarchical_level | 0.1401698  | 0.0107611  | 13.026  | < 2e-16 ***  |
| work_modeHybrid    | 0.1037704  | 0.0377119  | 2.752   | 0.00594 **   |
| work_modeRemote    | 0.1558910  | 0.0385129  | 4.048   | 5.21e-05 *** |
| ai_tools_dailyYes  | 0.1467178  | 0.0328724  | 4.463   | 8.17e-06 *** |
| Dead_c:Msup_c      | -0.0027014 | 0.0061002  | -0.443  | 0.65790      |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.406 on 9219 degrees of freedom

Multiple R-squared: 0.202, Adjusted R-squared: 0.2012

F-statistic: 233.4 on 10 and 9219 DF, p-value: < 2.2e-16

# Appendix 20: Moderation 1—Manager Support (H2)

> lm.beta(m\_h2)

Call:  
lm(formula = Burnout\_avg ~ Dead\_c \* Msup\_c + WHrs\_c + age + tenure +  
hierarchical\_level + work\_mode + ai\_tools\_daily, data = df)

Standardized Coefficients::

|                    | Dead_c          | Msup_c          | WHrs_c            | age           | tenure        |
|--------------------|-----------------|-----------------|-------------------|---------------|---------------|
| (Intercept)        |                 |                 |                   |               |               |
| NA                 | 0.1069980571    | -0.1156345797   | 0.3610516506      | -0.0009428147 | -0.0056532016 |
| hierarchical_level | work_modeHybrid | work_modeRemote | ai_tools_dailyYes | Dead_c:Msup_c |               |
| 0.1263716752       | 0.0323042677    | 0.0475186726    | 0.0415523256      | -0.0041238955 |               |

# Appendix 2: Moderation 1—Manager Support (H2)

```
> anova(m_h2_base, m_h2) #  $\Delta R^2$  – interaction adds variance?
```

Analysis of Variance Table

Model 1: Burnout\_avg ~ Dead\_c + Msup\_c + WHrs\_c + age + tenure + hierarchical\_level + work\_mode + ai\_tools\_daily

Model 2: Burnout\_avg ~ Dead\_c \* Msup\_c + WHrs\_c + age + tenure + hierarchical\_level + work\_mode + ai\_tools\_daily

|   | Res.Df | RSS   | Df | Sum of Sq | F      | Pr(>F) |
|---|--------|-------|----|-----------|--------|--------|
| 1 | 9220   | 18230 |    |           |        |        |
| 2 | 9219   | 18230 | 1  | 0.38778   | 0.1961 | 0.6579 |

JOHNSON-NEYMAN INTERVAL

When Msup\_c is **INSIDE** the interval [-11.678, 7.323], the slope of Dead\_c is  $p < .05$ .

Note: The range of observed values of Msup\_c is [-2.997, 3.003]

SIMPLE SLOPES ANALYSIS

Slope of Dead\_c when Msup\_c = -1.546491 (- 1 SD):

| Est.  | S.E.  | t val. | p     |
|-------|-------|--------|-------|
| 0.114 | 0.013 | 8.475  | 0.000 |

Slope of Dead\_c when Msup\_c = 1.546491 (+ 1 SD):

| Est.  | S.E.  | t val. | p     |
|-------|-------|--------|-------|
| 0.105 | 0.013 | 7.847  | 0.000 |

# Appendix 22: Moderation 2—Autonomy (H3)

Call:  
lm(formula = Burnout\_avg ~ Dead\_c \* Auto\_c + WHrs\_c + age + tenure +  
hierarchical\_level + work\_mode + ai\_tools\_daily, data = df)

Residuals:

| Min    | 1Q     | Median | 3Q    | Max   |
|--------|--------|--------|-------|-------|
| -4.311 | -1.085 | 0.015  | 1.061 | 3.983 |

Coefficients:

|                    | Estimate   | Std. Error | t value | Pr(> t )     |
|--------------------|------------|------------|---------|--------------|
| (Intercept)        | 3.3955902  | 0.0887548  | 38.258  | < 2e-16 ***  |
| Dead_c             | 0.1074021  | 0.0095687  | 11.224  | < 2e-16 ***  |
| Auto_c             | -0.0623155 | 0.0096059  | -6.487  | 9.19e-11 *** |
| WHrs_c             | 0.1090952  | 0.0029340  | 37.183  | < 2e-16 ***  |
| age                | -0.0001229 | 0.0023133  | -0.053  | 0.95764      |
| tenure             | -0.0029519 | 0.0060445  | -0.488  | 0.62531      |
| hierarchical_level | 0.1396089  | 0.0108269  | 12.895  | < 2e-16 ***  |
| work_modeHybrid    | 0.1086930  | 0.0379440  | 2.865   | 0.00419 **   |
| work_modeRemote    | 0.2054221  | 0.0393549  | 5.220   | 1.83e-07 *** |
| ai_tools_dailyYes  | 0.1453627  | 0.0330648  | 4.396   | 1.11e-05 *** |
| Dead_c:Auto_c      | -0.0017805 | 0.0060670  | -0.293  | 0.76917      |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.415 on 9219 degrees of freedom  
Multiple R-squared: 0.1923, Adjusted R-squared: 0.1915  
F-statistic: 219.5 on 10 and 9219 DF, p-value: < 2.2e-16

> lm.beta(m\_h3)

Call:  
lm(formula = Burnout\_avg ~ Dead\_c \* Auto\_c + WHrs\_c + age + tenure +  
hierarchical\_level + work\_mode + ai\_tools\_daily, data = df)

Standardized Coefficients::

| (Intercept)     | Dead_c            | Auto_c             | WHrs_c          |
|-----------------|-------------------|--------------------|-----------------|
| NA              | 0.1051157748      | -0.0621523853      | 0.3629710748    |
| age             | tenure            | hierarchical_level | work_modeHybrid |
| -0.0004973066   | -0.0045724087     | 0.1258660030       | 0.0338367218    |
| work_modeRemote | ai_tools_dailyYes | Dead_c:Auto_c      |                 |
| 0.0626167256    | 0.0411685490      | -0.0027484309      |                 |

> anova(m\_h3\_base, m\_h3)

Analysis of Variance Table

Model 1: Burnout\_avg ~ Dead\_c + Auto\_c + WHrs\_c + age + tenure + hierarchical\_level +  
work\_mode + ai\_tools\_daily

Model 2: Burnout\_avg ~ Dead\_c \* Auto\_c + WHrs\_c + age + tenure + hierarchical\_level +  
work\_mode + ai\_tools\_daily

|   | Res.Df | RSS   | Df | Sum of Sq | F      | Pr(>F) |
|---|--------|-------|----|-----------|--------|--------|
| 1 | 9220   | 18452 |    |           |        |        |
| 2 | 9219   | 18452 | 1  | 0.17237   | 0.0861 | 0.7692 |

# Appendix 23: Moderation 2—Autonomy (H3)

```
> lm.beta(m_h3)
```

Call:

```
lm(formula = Burnout_avg ~ Dead_c * Auto_c + WHrs_c + age + tenure +
    hierarchical_level + work_mode + ai_tools_daily, data = df)
```

Standardized Coefficients::

| (Intercept)     | Dead_c            | Auto_c             | WHrs_c          |
|-----------------|-------------------|--------------------|-----------------|
| NA              | 0.1051157748      | -0.0621523853      | 0.3629710748    |
| age             | tenure            | hierarchical_level | work_modeHybrid |
| -0.0004973066   | -0.0045724087     | 0.1258660030       | 0.0338367218    |
| work_modeRemote | ai_tools_dailyYes | Dead_c:Auto_c      |                 |
| 0.0626167256    | 0.0411685490      | -0.0027484309      |                 |

```
> anova(m_h3_base, m_h3)
```

Analysis of Variance Table

Model 1: Burnout\_avg ~ Dead\_c + Auto\_c + WHrs\_c + age + tenure + hierarchical\_level + work\_mode + ai\_tools\_daily

Model 2: Burnout\_avg ~ Dead\_c \* Auto\_c + WHrs\_c + age + tenure + hierarchical\_level + work\_mode + ai\_tools\_daily

|   | Res.Df | RSS   | Df | Sum of Sq | F      | Pr(>F) |
|---|--------|-------|----|-----------|--------|--------|
| 1 | 9220   | 18452 |    |           |        |        |
| 2 | 9219   | 18452 | 1  | 0.17237   | 0.0861 | 0.7692 |

```
> # — Simple slopes —————
> sim_slopes(m_h3, pred=Dead_c, modx=Auto_c,
+           modx.values="plus-minus", digits=3)
```

JOHNSON-NEYMAN INTERVAL

When Auto\_c is **INSIDE** the interval [-10.486, 7.713], the slope of Dead\_c is p < .05.

Note: The range of observed values of Auto\_c is [-3.002, 2.998]

SIMPLE SLOPES ANALYSIS

Slope of Dead\_c when Auto\_c = -1.569228 (- 1 SD):

| Est.  | S.E.  | t val. | p     |
|-------|-------|--------|-------|
| 0.110 | 0.013 | 8.173  | 0.000 |

Slope of Dead\_c when Auto\_c = 1.569228 (+ 1 SD):

| Est.  | S.E.  | t val. | p     |
|-------|-------|--------|-------|
| 0.105 | 0.014 | 7.742  | 0.000 |

```
>
```

```
> # — Interaction plot —————
```

```
> interact_plot(m_h3, pred=Dead_c, modx=Auto_c,
+             modx.values="plus-minus", plot.points=FALSE,
+             x.label="Deadline Pressure (centred)",
+             y.label="Burnout", legend.main="Autonomy")
```

```
> |
```

# Appendix 24: Moderation 2—AI tool (H4)

Call:  
lm(formula = Burnout\_avg ~ Dead\_c \* ai\_tools\_daily + WHrs\_c +  
age + tenure + hierarchical\_level + work\_mode + MgrSupp\_avg +  
Autonomy\_avg, data = df)

Residuals:

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -4.2360 | -1.0699 | 0.0108 | 1.0501 | 4.0087 |

|                          | Estimate   | Std. Error | t value  | Pr(> t ) |           |
|--------------------------|------------|------------|----------|----------|-----------|
| (Intercept)              | 4.115e+00  | 1.020e-01  | 40.325   | < 2e-16  | ***       |
| Dead_c                   | 1.272e-01  | 1.823e-02  | 6.981    | 3.13e-12 | ***       |
| ai_tools_dailyYes        | 1.446e-01  | 3.279e-02  | 4.409    | 1.05e-05 | ***       |
| WHrs_c                   | 1.087e-01  | 2.910e-03  | 37.371   | < 2e-16  | ***       |
| age                      | -6.607e-05 | 2.294e-03  | -0.029   | 0.97703  |           |
| tenure                   | -3.482e-03 | 5.995e-03  | -0.581   | 0.56136  |           |
| hierarchical_level       | 1.401e-01  | 1.074e-02  | 13.046   | < 2e-16  | ***       |
| work_modeHybrid          | 1.067e-01  | 3.763e-02  | 2.836    | 0.00458  | **        |
| work_modeRemote          | 2.018e-01  | 3.904e-02  | 5.168    | 2.42e-07 | ***       |
| MgrSupp_avg              | -1.173e-01 | 9.447e-03  | -12.420  | < 2e-16  | ***       |
| Autonomy_avg             | -6.223e-02 | 9.525e-03  | -6.533   | 6.78e-11 | ***       |
| Dead_c:ai_tools_dailyYes | -2.602e-02 | 2.136e-02  | -1.218   | 0.22307  |           |
| ---                      |            |            |          |          |           |
| Signif. codes:           | 0 '***'    | 0.001 '**' | 0.01 '*' | 0.05 '.' | 0.1 ' ' 1 |

Residual standard error: 1.403 on 9218 degrees of freedom  
Multiple R-squared: 0.2058, Adjusted R-squared: 0.2049  
F-statistic: 217.2 on 11 and 9218 DF, p-value: < 2.2e-16

```
> lm.beta(m_rq4_dead)

Call:
lm(formula = Burnout_avg ~ Dead_c * ai_tools_daily + WHrs_c +
  age + tenure + hierarchical_level + work_mode + MgrSupp_avg +
  Autonomy_avg, data = df)

Standardized Coefficients::
              (Intercept)              Dead_c              ai_tools_dailyYes              WHrs_c
                  NA              0.1245340852              0.0409460824              0.3618038186
                  age              tenure              hierarchical_level              work_modeHybrid
              -0.0002674317              -0.0053933331              0.1262666129              0.0332197988
              work_modeRemote              MgrSupp_avg              Autonomy_avg              Dead_c:ai_tools_dailyYes
              0.0614991707              -0.1153255813              -0.0620659272              -0.0217427626

> anova(m_rq4_base, m_rq4_dead)
Analysis of Variance Table

Model 1: Burnout_avg ~ Dead_c + WHrs_c + ai_tools_daily + age + tenure +
  hierarchical_level + work_mode + MgrSupp_avg + Autonomy_avg
Model 2: Burnout_avg ~ Dead_c * ai_tools_daily + WHrs_c + age + tenure +
  hierarchical_level + work_mode + MgrSupp_avg + Autonomy_avg
Res.Df  RSS Df Sum of Sq    F Pr(>F)
1    9219 18147
2    9218 18144   1    2.9223 1.4847 0.2231
> # 🔴 Dead_c:ai_tools_dailyYes = β₃ = THE moderation test
> # Positive β₃ → AI amplifies deadline→burnout (more cognitive load)
> # Negative β₃ → AI buffers deadline→burnout (efficiency gains)
>

> lm.beta(m_rq4_hrs)

Call:
lm(formula = Burnout_avg ~ WHrs_c * ai_tools_daily + Dead_c +
  age + tenure + hierarchical_level + work_mode + MgrSupp_avg +
  Autonomy_avg, data = df)

Standardized Coefficients::
              (Intercept)              WHrs_c              ai_tools_dailyYes              Dead_c
                  NA              0.3470907876              0.0410996969              0.1061307439
                  age              tenure              hierarchical_level              work_modeHybrid
              -0.0006285509              -0.0052476402              0.1262385423              0.0332420007
              work_modeRemote              MgrSupp_avg              Autonomy_avg              WHrs_c:ai_tools_dailyYes
              0.0611991729              -0.1156872100              -0.0618937779              0.0173801312

> anova(m_rq4_base, m_rq4_hrs)
Analysis of Variance Table

Model 1: Burnout_avg ~ Dead_c + WHrs_c + ai_tools_daily + age + tenure +
  hierarchical_level + work_mode + MgrSupp_avg + Autonomy_avg
Model 2: Burnout_avg ~ WHrs_c * ai_tools_daily + Dead_c + age + tenure +
  hierarchical_level + work_mode + MgrSupp_avg + Autonomy_avg
Res.Df  RSS Df Sum of Sq    F Pr(>F)
1    9219 18147
2    9218 18145   1    1.8274 0.9284 0.3353
> # 🔴 WHrs_c:ai_tools_dailyYes = β₃ = second moderation test
>
```

# Appendix 25: Moderation 2—AI tool (H4)

```
> # — Simple slopes —————
> sim_slopes(m_rq4_dead, pred = Dead_c, modx = ai_tools_daily,
+           digits = 3)
SIMPLE SLOPES ANALYSIS
```

Slope of *Dead\_c* when *ai\_tools\_daily* = No:

| Est.  | S.E.  | t val. | p     |
|-------|-------|--------|-------|
| 0.127 | 0.018 | 6.981  | 0.000 |

Slope of *Dead\_c* when *ai\_tools\_daily* = Yes:

| Est.  | S.E.  | t val. | p     |
|-------|-------|--------|-------|
| 0.101 | 0.011 | 9.103  | 0.000 |

Warning message:

Johnson-Neyman intervals are not available for factor predictors or moderators.

```
> # Shows slope of Deadline→Burnout separately for AI users vs non-users
>
```

SIMPLE SLOPES ANALYSIS

Slope of *WHrs\_c* when *ai\_tools\_daily* = No:

| Est.  | S.E.  | t val. | p     |
|-------|-------|--------|-------|
| 0.104 | 0.005 | 19.045 | 0.000 |

Slope of *WHrs\_c* when *ai\_tools\_daily* = Yes:

| Est.  | S.E.  | t val. | p     |
|-------|-------|--------|-------|
| 0.110 | 0.003 | 32.849 | 0.000 |

Warning message:

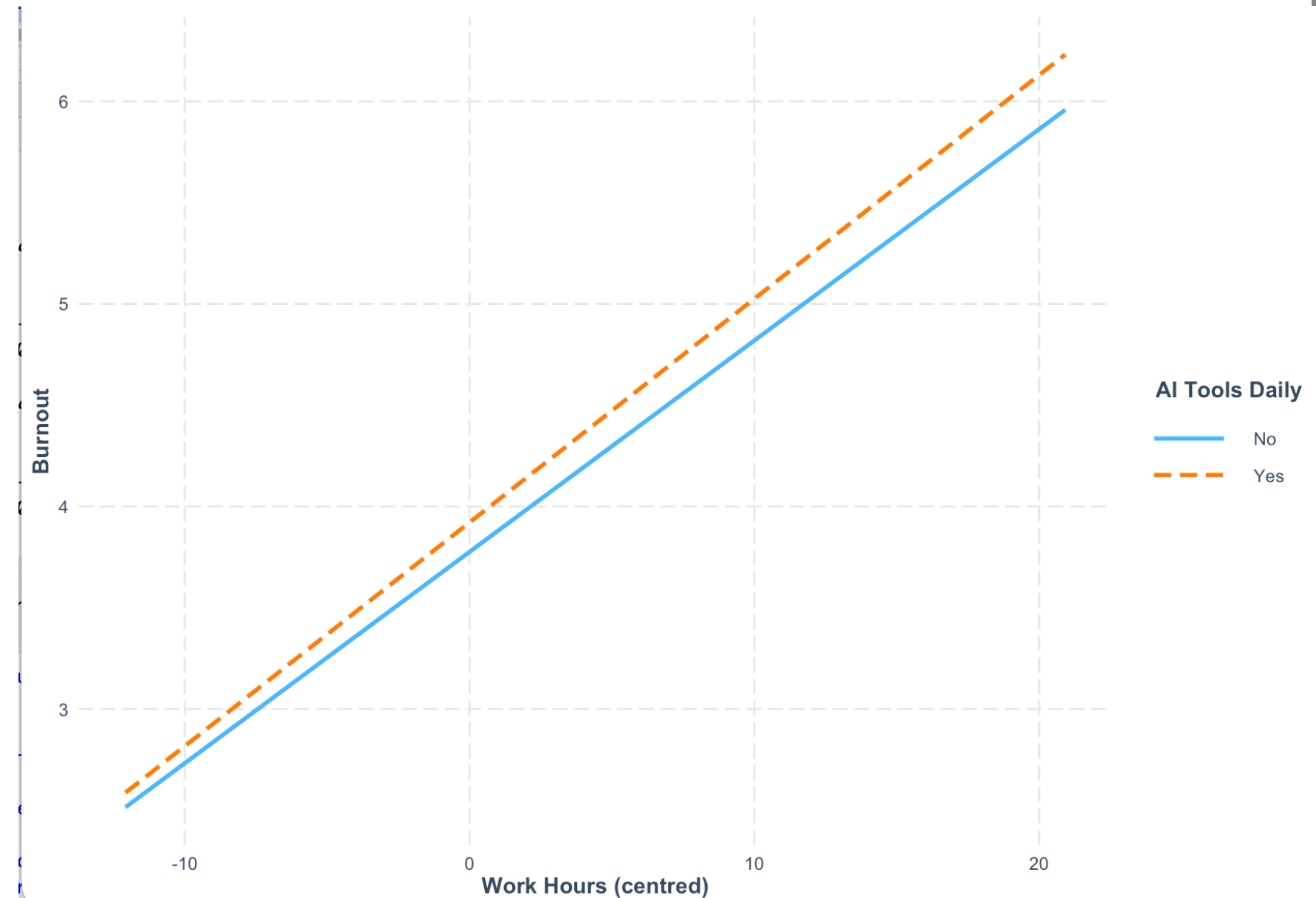
Johnson-Neyman intervals are not available for factor predictors or moderators.

```
> # Shows slope of WorkHours→Burnout separately for AI users vs non-users
>
```

```
> # — Interaction plots —————
```

```
> interact_plot(m_rq4_dead, pred = Dead_c, modx = ai_tools_daily,
+             x.label = "Deadline Pressure (centred)",
+             y.label = "Burnout", legend.main = "AI Tools Daily")
>
```

```
> interact_plot(m_rq4_hrs, pred = WHrs_c, modx = ai_tools_daily,
+             x.label = "Work Hours (centred)",
+             y.label = "Burnout", legend.main = "AI Tools Daily")
>
```



# Appendix 26: Diagnostics and Assumption Checks

```
> library(car)
>
> # — 1. Multicollinearity – VIF —————
> vif(m2)
```

|                     | GVIF     | Df | GVIF <sup>1/(2*Df)</sup> |
|---------------------|----------|----|--------------------------|
| Deadline_avg        | 1.057806 | 1  | 1.028497                 |
| work_hours_per_week | 1.183448 | 1  | 1.087864                 |
| Stress_avg          | 1.246236 | 1  | 1.116350                 |
| MgrSupp_avg         | 1.008422 | 1  | 1.004202                 |
| Autonomy_avg        | 1.049046 | 1  | 1.024229                 |
| age                 | 1.000734 | 1  | 1.000367                 |
| tenure              | 1.000472 | 1  | 1.000236                 |
| hierarchical_level  | 1.130029 | 1  | 1.063028                 |
| work_mode           | 1.050726 | 2  | 1.012447                 |
| ai_tools_daily      | 1.001747 | 1  | 1.000873                 |

```
> # — 3. Influential observations – Cook's Distance —————
> cooks_d <- cooks.distance(m2)
> threshold <- 4 / (nrow(df) - length(coef(m2)) - 1)
> sum(cooks_d > threshold)
[1] 380
> # Threshold ≈ .00043 at n=9,230 – some obs will exceed; check if results change
> # Sensitivity check: rerun m2 removing top 1% by Cook's D
> df_trim <- df[cooks_d < quantile(cooks_d, .99), ]
> m2_trim <- update(m2, data=df_trim)
> summary(m2_trim) # coefficients should be directionally consistent
```

```
Call:
lm(formula = Burnout_avg ~ Deadline_avg + work_hours_per_week +
    Stress_avg + MgrSupp_avg + Autonomy_avg + age + tenure +
    hierarchical_level + work_mode + ai_tools_daily, data = df_trim)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-3.7477 -0.8504 -0.0029  0.8308  3.9277
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -1.0301858   0.1394693  -7.386 1.64e-13 ***
Deadline_avg  -0.0028825   0.0081233  -0.355 0.722714
work_hours_per_week  0.0661956   0.0025295  26.170 < 2e-16 ***
Stress_avg     0.5322441   0.0085771  62.054 < 2e-16 ***
MgrSupp_avg    -0.0776205   0.0079037  -9.821 < 2e-16 ***
Autonomy_avg   -0.0420495   0.0079389  -5.297 1.21e-07 ***
age            0.0010179   0.0019121   0.532 0.594522
tenure         -0.0001117   0.0050802  -0.022 0.982463
hierarchical_level  0.0355779   0.0091454   3.890 0.000101 ***
work_modeHybrid  0.0896791   0.0313289   2.863 0.004213 **
work_modeRemote  0.1276375   0.0325541   3.921 8.89e-05 ***
ai_tools_dailyYes  0.1079335   0.0273122   3.952 7.81e-05 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.161 on 9125 degrees of freedom
Multiple R-squared:  0.4502,    Adjusted R-squared:  0.4495
F-statistic: 679.2 on 11 and 9125 DF,  p-value: < 2.2e-16
```